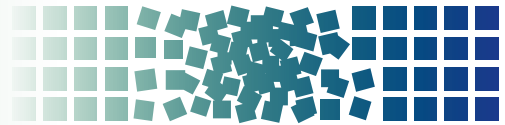


# 50 Cancer of the Uterine Body



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## ENDOMETRIAL CARCINOMA

### Epidemiology

Endometrial cancer is the most common gynecologic malignancy and the fourth most frequently diagnosed cancer in women in the United States. The American Cancer Society estimates there would be 65,950 new cases and 12,550 deaths in 2022.<sup>1</sup> The average age of diagnosis for endometrial cancer is 63 years, yet there seems to be a sustained increase in number of cases for women under the age of 50 years old.<sup>2</sup> One of every 35 women (2.8%) is likely to develop endometrial cancer during her lifetime. The death rate from this disease seems to be on the rise, by about 2% per year,<sup>1</sup> and racial disparity and socioeconomic and geographical differences are important determinants in the incidence and mortality associated with endometrial cancer.<sup>3</sup> In contrast to endometrioid adenocarcinomas, the incidence of nonendometrioid subtypes have increased by approximately 2.9% per year from 2000 to 2015. Additionally, higher rates of nonendometrioid subtypes over this study period have been seen in black women (25.9 per 100,000 women) compared to white women (11.4), Hispanic women (10.1), and Asian/Pacific Islander women (7.5).<sup>3</sup>

The exact etiology of endometrial cancer remains unknown, but chronic unopposed estrogenic stimulation is considered the main risk factor. The normal endometrium is a hormonally responsive tissue; estrogenic stimulation produces cellular growth and glandular proliferation, which is cyclically balanced by the maturational effects of progesterone. Because menarche and menopause are commonly associated with absent or irregular ovulation, women who experience *early menarche* or *late menopause* are more likely to have additional estrogenic exposure and thus a higher risk of developing endometrial cancer. *Nulliparity* is also associated with increased risk of endometrial cancer, due in part to anovulatory menstrual cycles. *Obesity* is one of the strongest risk factors for endometrial cancer and accounts for about 57% of the cases in the USA. It is estimated that for every 5-unit increase in the BMI, the risk of developing endometrial cancer increases by about 50%.<sup>2</sup> Morbidly obese women are at greatest risk of endometrial cancer, presumably because their adipocytes convert androstenedione of adrenal origin to estrone, a weak circulating estrogen. Obesity-associated hyperglycemia, insulin resistance, and resulting abnormalities in IGF-1 signaling can activate the mammalian target of rapamycin (mTOR) pathway and stimulate increased cell proliferation. Associated inflammation and oxidative stress, as well as alterations in cytokines, steroid hormones, and adipokine pathophysiology, can promote endometrial oncogenesis in obese women.<sup>4</sup> *Non-insulin-dependent diabetes mellitus* and *hypertension* also increase the risk of endometrial cancer, in large part secondary to obesity. The use of *estrogen-only hormone replacement therapy* and *sequential oral contraceptives* significantly increases endometrial cancer risk, whereas combined preparations (i.e., that contain progestogen) have a protective effect.<sup>2</sup>

The use of *tamoxifen* in patients with breast cancer has been associated with increased risk of endometrial cancer. In premenopausal women, tamoxifen has an antiestrogenic effect, but in postmenopausal women, it has a weak estrogenic effect because of the upregulation of estrogen receptors. It is estimated that the use of tamoxifen doubles the risk of developing endometrioid as well as nonendometrioid endometrial cancer. This risk is four times higher when tamoxifen is used for more than 5 years.<sup>2</sup>

Approximately 3% of all endometrial cancers (9% in those under the age of 50) are associated with *Lynch syndrome*, or hereditary nonpolyposis colorectal cancer.<sup>2</sup> Lynch syndrome is an autosomal dominant inherited cancer susceptibility syndrome caused by a germline mutation in one of the DNA mismatch repair genes (*MSH2*, *MLH1*, *MSH6*, *PMS2*). Women with this syndrome have a 40% to 60% risk of endometrial cancer, which equals or exceeds their risk of colorectal cancer. The estimated lifetime risk for endometrial cancer ranges from 40% to 60% for *MLH1* or *MSH2* mutation carriers, 16% to 20% for *MSH6* carriers, and 15% for *PMS2* carriers. These endometrial cancers are diagnosed at a mean age of 50 years, as opposed to 61 years in sporadic cases,<sup>5</sup> and more frequently involve the lower uterine segment. In terms of histology, endometrioid is the most common, but others, such as clear cell carcinoma, serous carcinoma, and carcinosarcoma, have been reported in patients with Lynch syndrome. Cowden syndrome is associated with an increased risk of endometrial, breast, and thyroid cancers. The lifetime risk of endometrial cancer in Cowden syndrome is as high as 28%.<sup>6</sup>

### Natural History and Routes of Spread

The most common presentation for endometrial cancer is postmenopausal vaginal bleeding, which is reported by 80% to 90% of patients. The incidence of endometrial cancer in women presenting with postmenopausal bleeding, however, is only 10% to 15%. This incidence could vary from 1% up to 25% depending on patient age, the number of vaginal bleeding episodes, and the presence of other risk factors such as diabetes and high body mass index. Other patterns of presentations include vaginal discharge, abnormal Papanicolaou smear, or thickened endometrium on routine transvaginal ultrasound. Presenting with urinary or rectal bleeding, pain, lower extremity lymphedema, ascites, and cough and/or hemoptysis is rare and often indicative of advanced disease.

The uterus has a rich and complex lymphatic network. The main body of the uterus drains via lymphatic trunks that condense in the parametria and end up in pelvic nodes. Channels draining the superior portion of the fundus and fundal uterine serosa parallel the ovarian vessels and empty into the para-aortic lymph nodes in the upper abdomen. A few small lymphatic vessels course through the round ligaments to the superficial inguinal nodes. Invasion into adjacent organs such as bladder and rectum may occur, although this does not occur frequently. Tumor cells may gain access to the peritoneal cavity, at times leading to peritoneal implants, similar to ovarian cancers. Sites of distant spread include lungs, liver, and bone.

## Diagnosis and Pretherapy Evaluation

Office endometrial biopsy is the preferred approach to establish the diagnosis of endometrial cancer. Its sensitivity in detecting endometrial cancer in postmenopausal women is close to 99%, compared to about 91% in premenopausal women, and its specificity is >98% for both groups. However, if symptoms persist, the biopsy sampling is inadequate, or the patient is being considered for conservative fertility-sparing approaches, a dilatation and curettage should be performed instead. To confirm a diagnosis of endometrial cancer, a tissue diagnosis is required and must not be substituted by imaging studies. *Transvaginal ultrasonography* may be a useful imaging modality, particularly in patients who are medically compromised such that obtaining a tissue sample is not feasible. Normal postmenopausal endometrium looks thin and homogeneously hyperechoic, but in endometrial cancer, it becomes thickened and heterogeneous with hyperplasia, polyps, and tumors. The consensus statement from the Society of Radiologists in Ultrasound has defined an endometrial thickness of  $\geq 5$  mm as being abnormal.<sup>7</sup> If the thickness of the endometrium is <5 mm, the risk of endometrial cancer is minimal; the false-negative rate is approximately 4%.

*Computed tomography (CT) imaging* is not very useful in determining myometrial invasion (sensitivity 83%, specificity 42%) or in assessing cervical stromal invasion.<sup>8</sup> It is helpful, however, in assessing potential spread, especially for those with higher grade histology and high-risk histologic subtypes such as serous or clear cell carcinoma as opposed to those with endometrial hyperplasia or endometrioid low-grade carcinoma.

*Magnetic resonance imaging (MRI)*, especially dynamic contrast-enhanced MRI, is very useful in detecting myometrial invasion, with an accuracy of 98%.<sup>8</sup> In patients with suspected cervical involvement, preoperative MRI may also help determine whether the uterine tumor involves the lower uterine segment or truly extends into the cervix. Gross cervical involvement in endometrial cancer is rare but important to document because it could influence surgical treatment (radical hysterectomy as opposed to simple hysterectomy). *Positron emission tomography (PET) plus CT* is of little benefit in assessing the primary tumor extension, of moderate benefit in detecting nodal metastasis with an overall diagnostic accuracy of 88%, and even greater benefit in detecting distant metastasis and relapse with an overall diagnostic accuracy of 93%.<sup>9</sup> *Serum cancer antigen (CA) 125* levels may be a predictor of extrauterine disease. In a study of 214 endometrial cancer patients, serum CA 125 was found to be an independent risk factor for pelvic and para-aortic lymph node metastasis.<sup>10</sup> Elevated levels of CA 125 may also assist in predicting treatment response or in posttreatment surveillance.

## Pathology

Endometrial *hyperplasia* is the precursor lesion of the most common endometrial cancer, endometrioid adenocarcinoma. The World Health Organization classifies endometrial hyperplasia into four subtypes: simple hyperplasia, complex hyperplasia, simple atypical hyperplasia, and complex atypical hyperplasia. The risk of progressing to endometrial cancer is approximately 1% for simple hyperplasia but increases up to approximately 29% for complex hyperplasia associated with atypia. In a prospective trial conducted by the Gynecologic Oncology Group (GOG), among patients with atypical hyperplasia of the uterus who underwent immediate hysterectomy, the rate of underlying concurrent carcinoma in the uterus was 42.6%.<sup>11</sup> The standard recommended treatment for atypical hyperplasia of the uterus is hysterectomy; however, for patients who desire future fertility or have an absolute contraindication to surgery, an alternative could be oral progestin or a progestin containing intrauterine device (IUD). Such a conservative approach needs to be used with caution and requires close monitoring.<sup>2</sup>

TABLE 50.1

### Pathologic Classification of Endometrial Cancers

|                                              |
|----------------------------------------------|
| Endometrioid adenocarcinoma                  |
| Not otherwise specified                      |
| Adenocarcinoma with squamous differentiation |
| Secretory adenocarcinoma                     |
| Ciliated carcinoma                           |
| Villoglandular                               |
| Serous                                       |
| Clear cell carcinoma                         |
| Undifferentiated carcinoma                   |
| Mucinous carcinoma                           |
| Squamous cell carcinoma                      |
| Transitional cell carcinoma                  |
| Mixed cell type                              |

The most common histologic subtype of endometrial cancer is *endometrioid*. The grading is primarily driven by architecture (i.e., the proportion of solid masses of tumor cells relative to well-defined glands). Grade 1 is an endometrioid cancer in which <5% of the tumor growth is in solid sheets. Grade 2 is an adenocarcinoma in which 6% to 50% of the tumor is composed of solid sheets of cells. Grade 3 occurs when >50% of the tumor is made up of solid sheets. There are several subtypes of endometrioid adenocarcinoma, as shown in Table 50.1. *Serous carcinoma*, also known as uterine papillary serous cancer, is seen in approximately 10% of endometrial cancers. The presence of marked cellular atypia in addition to papilla is what distinguishes serous carcinoma from others. This is an aggressive subtype with a high propensity for early lymphatic and intraperitoneal dissemination. Other histologies include clear cell carcinoma (<5% of endometrial cancer) and undifferentiated carcinomas.

## Prognostic Factors

Adverse prognostic factors in endometrial cancer that are included in the staging system include depth of myometrial invasion, cervical stromal involvement, adnexal involvement, pelvic/para-aortic node involvement, extension into bladder or rectum, and distant spread. Stage correlates with risk of recurrence, as confirmed by an NRG Oncology/GOG observational study, where the rate of recurrence was 9.8% among 3,484 Stage I patients, 21.6% for 361 Stage II patients, and 38.9% for 853 Stage III patients.<sup>12</sup> Other non-staging-related prognostic factors include age, race, grade, histologic type, and lymphovascular invasion (LVI). The prevalence of these prognostic factors, however, is stage dependent, making the assessment of their independent impact somewhat challenging.

The adverse impact of *older age* is often explained away by pointing out that older patients tend to present with aggressive histologies and more advanced disease and are generally treated less aggressively. Age 60 years and older, however, has been shown to be predictive of locoregional recurrence (hazard ratio [HR], 3.9;  $P = .0017$ ) and death (HR, 2.66;  $P = .01$ ) in a randomized trial limited to Stage I disease, excluding patients with deep myometrial invasion and grade 3 tumors.<sup>13</sup> Further, the adverse impact of older age persists even when they are treated as aggressively as younger patients.<sup>14</sup> Similarly, the influence of race (African Americans and Hispanics have worse outcomes compared to Caucasians) is often attributed to advanced stage, and aggressive histologies. However, NRG Oncology data<sup>12</sup> seem to show that the impact of race persists even after adjusting for such confounding factors, making the case for exploring the potential impact of race on tumor biology as well as access to appropriate care.<sup>2,15</sup>

Although prognosis may not differ significantly between grade 1 and 2 patients, those with *grade 3* clearly have worse outcomes.<sup>16</sup>

In Postoperative Radiation Therapy in Endometrial Carcinoma (PORTEC)-1 trial,<sup>13</sup> which was limited to Stage I disease, grade 3 was associated with increased risk of locoregional recurrence (HR, 3.35;  $P = .0003$ ) and death (HR, 7.3;  $P < .0001$ ).

Histology can influence outcome even within early-stage disease. An NRG Oncology/GOG study demonstrated that survival and recurrence rates for patients with clear cell carcinoma were similar to those for endometrioid grade 3 disease but were significantly better than those for serous endometrial cancer.<sup>17</sup>

LVI is associated with metastasis. Among the 22.7% of patients in the NRG Oncology/GOG-210 study whose cancers displayed LVI, the rate of pelvic node metastasis was 37.5%, compared with 4.1% among those without, and the rate of para-aortic metastasis was 23.8% (vs 2.3%).<sup>18</sup> This translates into more frequent relapses, including vaginal recurrences.<sup>19</sup> The prognostic significance of LVI is dependent on its extent and the presence of other prognostic factors.<sup>20,21</sup>

To incorporate the impact of several prognostic factors on outcome, investigators at Memorial Sloan Kettering Cancer Center (MSKCC) developed a nomogram for predicting overall survival (OS) of women with endometrial cancer ( $n = 1,735$ ) following primary therapy.<sup>22</sup> Five prognostic factors (age, grade, histologic type, number of lymph nodes removed, and International Federation of Gynecology and Obstetrics [FIGO] 1988 surgical stage) were used to predict OS with high concordance probability. PORTEC also developed a similar nomogram, which only covers early-stage disease.<sup>23</sup>

## Endometrial Cancer Molecular Classification

It has been long recognized that there are two different pathogenetic types of endometrial carcinoma. Type I disease arises in women with obesity and signs of hyperestrogenism, such as anovulatory uterine bleeding, infertility, late onset of menopause, and hyperplasia of the endometrium. These tumors tend to be well to moderately differentiated, with minimal invasion, and carry a good

prognosis. Type II arises in women who lack signs of hyperestrogenism and tend to be older. These tumors are often poorly differentiated, with deep myometrial invasion and frequent occurrence of extrauterine spread.<sup>2</sup>

Remarkably, this clinical observation has been confirmed and further refined by The Cancer Genome Atlas (TCGA) Research Network, providing the first comprehensive proteomic and transcriptomic analysis of endometrial cancer resulting in a new classification of endometrial cancer into the following four different subtypes: the polymerase  $\epsilon$  (POLE)-ultramutated, MSI-hypermutated (MSI-H), copy number low, and copy number high endometrial carcinoma.<sup>24</sup> POLE-ultramutated endometrial cancers (approximately 10% of endometrioid histology) are microsatellite stable (MSS) tumors characterized by a very high mutation rate with frequent mutations in *PTEN*, *PIK3R1*, *PIK3CA*, and *RAS*. MSI-H tumors show microsatellite instability (MSI) and high mutation rates due to defects in mismatch repair (MMR) genes (*MLH1*, *MSH2*, *MSH6*, and *PMS2*), both in sporadic and hereditary endometrial cancer. *PTEN* mutations, alterations in the *PTEN*-*PIK3CA* pathway, and *KRAS* mutations are also frequent in this subtype. The copy number low subgroup is MSS and is characterized by a low mutation rate and is associated with frequent *PTEN* (77%) and *PIK3CA* (53%) mutations, as well as WNT- $\beta$ -catenin pathway alterations (52%). The copy number high subgroup includes mainly serous carcinomas and grade 3 endometrioid adenocarcinomas (25%, called "serous like"). *TP53* is commonly mutated (>90% of cases), and *ERBB2* amplification is detected in approximately 25% of the serous carcinomas. (Fig. 50.1).

TCGA endometrial cancer classification entails methodology that requires fresh-frozen material and expensive complex sequencing platforms. In an effort to decrease cost and technical challenges associated with these methods, a simplified molecular classifier, ProMisE, has been developed,<sup>25</sup> which identifies four molecular subtypes of EC: MMRd, DNA POLE, p53aberrant, and p53 wild type. Cases lacking adequate information to further classify are designated as No Specific Molecular Profile (NSMP).<sup>26</sup> While determining MMR and p53 status via immunohistochemistry (IHC)

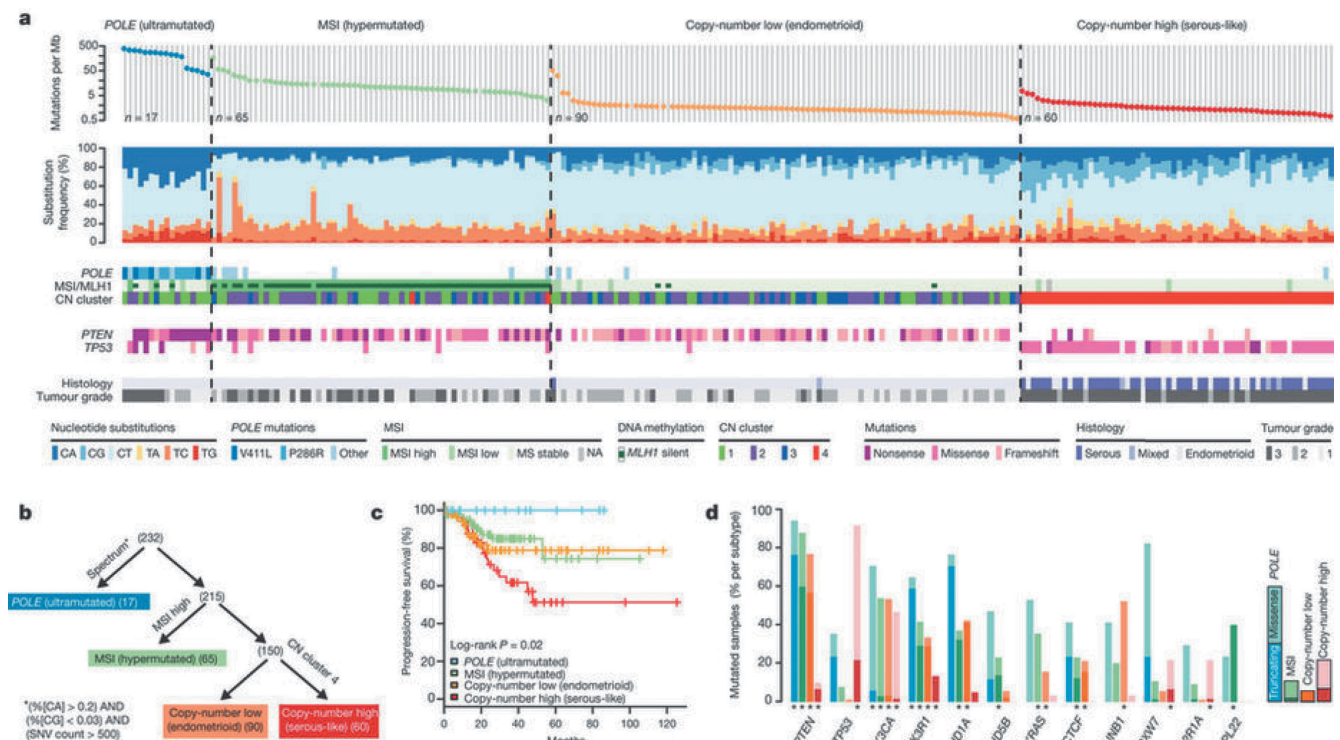


Figure 50.1 Genomic landscape of endometrial cancer.

is widely available, sequencing POLE is not currently broadly utilized. IHC analyses as part of endometrial cancer diagnosis should be universally implemented to guide treatment of patients.

Several studies have shown the four molecular subtypes of endometrial cancer delineated by the ProMisE Classifier are associated with differential survival<sup>25–27</sup> with the p53abn group having the poorest prognosis, with a risk of death or progressive/recurrent disease three- to fivefold higher compared to the p53wt group and twofold higher when adjusted for clinicopathological features. The MSI group has a 1.5- to twofold increased risk of death compared to the p53wt group, which was nonsignificant after adjusting for clinicopathological features. The POLE-mutant group is least impacted by clinicopathological features and is associated with the most favorable prognosis.

Genomic analysis of clear cell carcinoma of the uterus found four genomic groups, similar to those in endometrioid carcinomas: POLE, mismatch repair-deficient tumors, copy number low, and copy number high.<sup>28</sup>

## Staging

According to the NRG Oncology/GOG-210 study,<sup>18</sup> the extent of disease was as follows: 24.8% of patients had disease limited to the endometrium, 38.4% had disease limited to the inner half of the myometrium, 11.2% had disease that involved only the outer half of the myometrium, 2.3% had disease that extended only to the endocervical gland, 4.4% had tumor that extended to the cervical stroma, 8.6% had adnexal metastasis, 9.8% had pelvic node metastasis, 5.1% had para-aortic metastasis, and 10.7% had other extrauterine metastasis.

In 2009, the FIGO Committee on Gynecologic Oncology (Table 50.2) revised the 1988 staging for endometrial cancer.<sup>29</sup> The discriminating power of the 2009 FIGO staging system in Stage I

disease is debatable. A study from MSKCC of 1,307 patients with FIGO 1988 Stage I disease showed that the revised system for Stage I did not improve its predictive ability over the 1988 system.<sup>30</sup> Endocervical glandular involvement is not considered in staging; only patients with cervical stromal invasion are classified as Stage II. Endometrial cancers with parametrial extension are now considered Stage IIIB. Patients with Stage IIIC are subdivided into IIIC1 if pelvic nodes are involved and IIIC2 if para-aortic nodes are involved. In the era of sentinel mapping, the extent of nodal involvement has been redefined: macro-metastasis is defined as >2.0 mm, micro-metastasis as >0.2 mm or >200 cells, and isolated tumor cells (ITCs) as a cluster of cells ≤0.2 mm or ≤200 cells.<sup>2</sup>

Although positive peritoneal cytology, seen in 8.9% of patients in the NRG Oncology/GOG-210 study,<sup>18</sup> is no longer considered in staging, it has prognostic impact in patients with other extra-uterine features. Milgrom et al.<sup>31</sup> reported on 196 patients with Stage III (2009 FIGO staging system) endometrial cancer, where positive peritoneal cytology was associated with an increased hazard for relapse (HR, 2.3;  $P = .002$ ) and death (HR, 2.9;  $P < .001$ ). This characteristic seems to have little prognostic significance in early-stage, low-grade endometrioid carcinoma.<sup>32</sup>

Parametrial extension is considered IIIB. Patients with Stage IIIC are subdivided into IIIC1 if pelvic nodes are involved and IIIC2 if para-aortic nodes are involved.

## Treatment of Early-Stage (Stage I and II) Disease

### Surgery

The main treatment is simple hysterectomy and bilateral salpingo-oophorectomy (BSO), peritoneal cytology, and some form of regional lymph node assessment. Salpingo-oophorectomy is recommended because 8.6% of patients with endometrial cancer may have adnexal involvement.<sup>18</sup> This procedure also eliminates synchronous ovarian cancer, found in 5% of all women with endometrial cancer; among women younger than 50 years old, the rate could be close to 10%. Differentiating between metastasis to ovary as opposed to synchronous ovarian cancer is not always easy. Genetic profiling may represent a powerful tool in clinical practice to distinguish between metastatic and dual primaries in patients with simultaneous ovarian and endometrial cancer and to predict disease outcome.<sup>33</sup> The 2009 FIGO staging system requires that peritoneal cytology be obtained and reported separately from stage.

Simple hysterectomy can be performed via an open approach (i.e., *total abdominal hysterectomy* [TAH]/BSO) or a minimally invasive approach (i.e., *laparoscopic-assisted hysterectomy*/BSO or *robotic-assisted hysterectomy*/BSO). In a prospective GOG trial, patients with clinical Stage I to occult IIA uterine cancer were randomly assigned to laparoscopy ( $n = 1,696$ ) or open laparotomy ( $n = 920$ ), including hysterectomy, salpingo-oophorectomy, pelvic cytology, and pelvic and para-aortic lymphadenectomy.<sup>34</sup> Laparoscopy was associated with fewer moderate to severe postoperative adverse events than laparotomy (14% vs 21%, respectively;  $P < .0001$ ) and lower rates of extended hospitalization (>2 days; 52% vs 94%, respectively;  $P < .0001$ ). The conversion rate to laparotomy was 25.8%. The estimated 5-year OS was the same (89.8% for both procedures).<sup>35</sup> The specific method of minimally invasive surgery, whether conventional laparoscopy or robotic surgery, should be tailored to patient selection, surgeon ability, and equipment availability. Because minimally invasive surgery shortens hospital stays and reduces subsequent emergency room visits and readmission, it is likely to reduce the cost of endometrial cancer treatment.<sup>36</sup>

Regarding regional lymph node assessment, the options range from removing only grossly suspicious nodes to routine lymphadenectomy for all patients. Based on NRG Oncology/GOG-120 surgical-pathologic assessment, the rate of positive pelvic lymph nodes ranges from as low as 0.8% for grade 1 endometrioid

**TABLE 50.2**

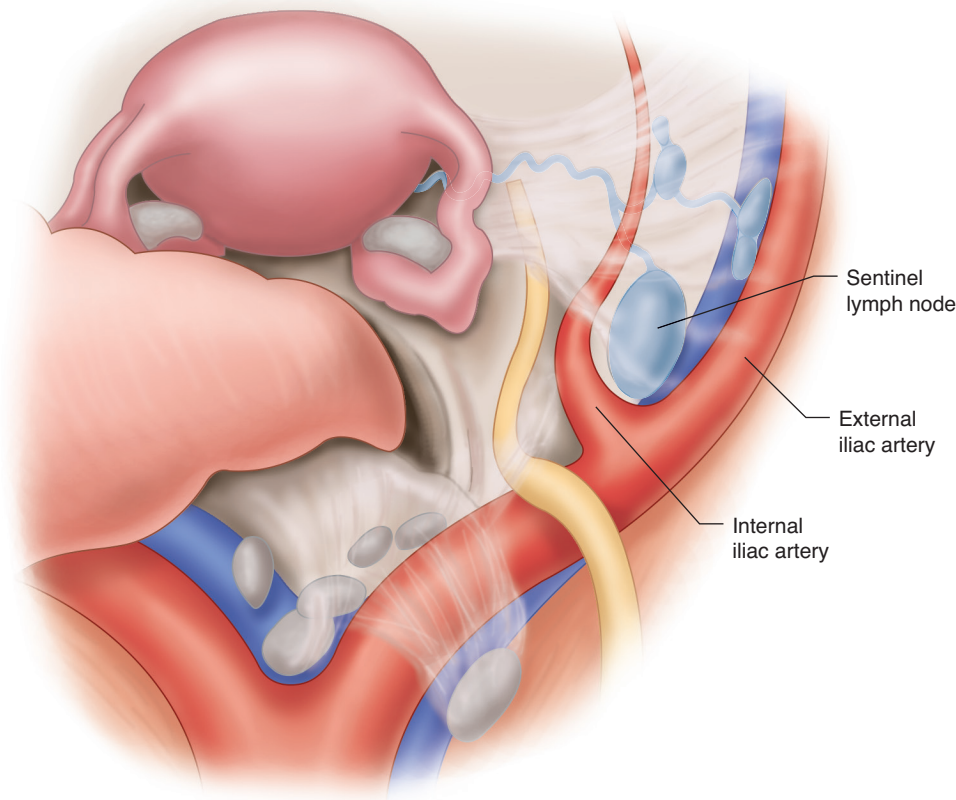
### Revised International Federation of Gynecology and Obstetrics Staging for Endometrial Cancer

|                          |                                                                                      |
|--------------------------|--------------------------------------------------------------------------------------|
| Stage I <sup>a</sup>     | Tumor confined to the corpus uteri                                                   |
| IA <sup>a</sup>          | No or less than half myometrial invasion                                             |
| IB <sup>a</sup>          | Invasion equal to or more than half of the myometrium                                |
| Stage II <sup>a</sup>    | Tumor invades cervical stroma but does not extend beyond the uterus <sup>b</sup>     |
| Stage III <sup>a,c</sup> | Local and/or regional spread of the tumor                                            |
| IIIA <sup>a</sup>        | Tumor invades the serosa of the corpus uteri and/or adnexae <sup>a</sup>             |
| IIIB <sup>a</sup>        | Vaginal and/or parametrial involvement <sup>a</sup>                                  |
| IIIC <sup>a</sup>        | Metastases to pelvic and/or para-aortic lymph nodes <sup>a</sup>                     |
| IIIC1                    | Positive pelvic nodes                                                                |
| IIIC2                    | Positive para-aortic lymph nodes with or without positive pelvic nodes               |
| Stage IV <sup>a</sup>    | Tumor invades bladder and/or bowel mucosa, and/or distant metastases                 |
| IVA <sup>a</sup>         | Tumor invasion of bladder and/or bowel mucosa                                        |
| IVB <sup>a</sup>         | Distant metastases, including intra-abdominal metastases and/or inguinal lymph nodes |

<sup>a</sup>Grade 1, grade 2, or grade 3.

<sup>b</sup>Endocervical glandular involvement only should be considered as stage I and no longer as stage II.

<sup>c</sup>Positive cytology has to be reported separately without changing the stage. With permission from Pecorelli S. Revised FIGO staging for carcinoma of the vulva, cervix, and endometrium. *Int J Gynecol Obstet* 2009;105:103–104.



**Figure 50.2** Diagrammatic illustration of the location of the sentinel lymph nodes (blue) in relationship to vessels between internal and external iliac arteries).

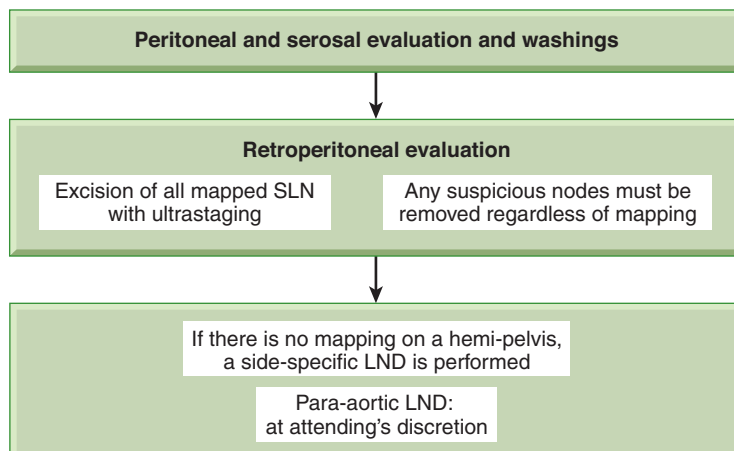
carcinoma without myometrial invasion to as high as 44.3% for patients with serous histology and deep (>50%) myometrial invasion.<sup>18</sup> Two prospective randomized trials assessed the benefit of *pelvic lymphadenectomy* in clinical Stage I endometrial cancer (para-aortic node sampling was optional); both found that pelvic lymphadenectomy did not improve OS.<sup>37,38</sup>

A middle ground approach between removing only grossly suspicious nodes and routine lymphadenectomy is *sentinel lymph node (SLN) mapping* (Fig. 50.2). Two prospective studies demonstrated that successful mapping is achievable in 86% to 89% of cases, with a sensitivity of 84% to 97.2% and a negative predictive value of 97% to 99.6%. Reassuringly, the rate of positive nodes is 12% to 17%, slightly higher than that after lymphadenectomy.<sup>39,40</sup> At MSKCC, an SLN mapping algorithm is used to further refine the utility of this technique.<sup>41</sup> SLN mapping is for normal-appearing nodes; any grossly enlarged or suspicious nodes must be evaluated irrespective of mapping. In fact, such involved nodes often do not absorb the injected

dye or radiocolloid (i.e., they do not map). Similarly, failed mapping on a hemipelvis should be accompanied by side-specific lymphadenectomy to include the iliac and obturator nodes (Fig. 50.3). This algorithm reduced the false-negative rate from 15% to 2%.

With such encouraging results, SLN mapping is gaining acceptance in the staging of endometrioid adenocarcinoma and to some extent in nonendometrioid histologies. In a prospective study of 123 patients with grade 3 endometrioid histology, serous carcinoma, clear cell carcinoma, or carcinosarcoma, all patients underwent SLN mapping followed by lymphadenectomy. The SLN detection rate was 89%, with a sensitivity of 95% and false-negative rate of 5% (1 of 20 patients). The rate of positive nodes was 23%; only one patient had negative SLNs and a positive non-SLN.<sup>42</sup>

In summary, minimally invasive simple hysterectomy and BSO (laparoscopic or robotic), peritoneal cytology, and SLN mapping seem to provide the highest prognostic and therapeutic yield while maintaining a low morbidity profile.



**Figure 50.3** Sentinel lymph node mapping algorithm. SLN, sentinel lymph node; LND, lymphadenectomy.

TABLE 50.3

### Treatment Recommendations for Early-Stage (FIGO 2009) Endometrioid Adenocarcinoma of the Uterus

|                                                  | Grade 1 or 2             | Grade 3              |
|--------------------------------------------------|--------------------------|----------------------|
| IA                                               | Observation <sup>a</sup> | Observation vs IVRT  |
| IB                                               | IVRT                     | IMRT                 |
| II <sup>b</sup> (<50% cervical stromal invasion) | IVRT vs IMRT             | IMRT vs IVRT plus CT |
| II (>50% cervical stromal invasion)              | IMRT                     | IMRT vs IVRT plus CT |

<sup>a</sup>Consider IVRT for patients with <50% myometrial invasion who are also aged 60 years and older or have lymphovascular invasion.

<sup>b</sup>Patients with endocervical gland involvement, although no longer considered stage II, should be treated in a similar fashion to this subgroup of stage II patients.

FIGO, International Federation of Gynecology and Obstetrics; IVRT, intravaginal radiation; IMRT, intensity-modulated radiation therapy; CT, chemotherapy, generally paclitaxel/carboplatin.

## Radiation

Indications and types of radiation for early-stage endometrioid adenocarcinoma are shown in Table 50.3. For patients with *Stage IA, grade 1 and 2* endometrioid adenocarcinoma (no or <50% myometrial invasion), adjuvant radiotherapy (RT) is not routinely recommended. In a Swedish trial, 645 such patients were randomized after surgery to observation ( $n = 326$ ) or intravaginal RT ( $n = 319$ ). Surgery consisted of TAH/BSO (including laparoscopic procedures), pelvic washing, and removal of enlarged nodes. The rates of vaginal and pelvic recurrence did not differ between the observation and intravaginal RT arms (vaginal recurrence, 3.1% vs 1.2%,  $P = .114$ ; pelvic recurrence, 0.9% vs 0.3%,  $P = .326$ ), nor did OS. Patients who developed recurrence were significantly older than those who did not (mean age, 68.6 vs 62.6 years;  $P = .018$ ).<sup>43</sup> In addition to older age, LVI is associated with a greater risk of vaginal relapse among patients with Stage I endometrial cancer, even with minimal myometrial invasion (7% vs 3%,  $P = .02$ ).<sup>19</sup> Therefore, although observation is a valid option for patient with grade 1 or 2 disease and <50% myometrial invasion, MSKCC recommends intravaginal RT for patients with myometrial invasion that is <50% but who are 60 years and older or have LVI.

For patients with *Stage IA, grade 3* endometrioid adenocarcinoma, those without myometrial invasion may be observed or treated with postoperative intravaginal RT, whereas patients with <50% invasion are often treated with adjuvant radiation. In the PORTEC-1 trial, 715 patients with Stage I endometrial cancer (excluding patients with <50% myometrial invasion and grade 1 and patients with >50% invasion and grade 3) were randomized after TAH/BSO to observation or pelvic RT.<sup>13</sup> In the subset of patients with <50% myometrial invasion and grade 3 endometrioid tumors ( $n = 37$ ), the 5-year vaginal recurrence rate was lower for those treated with pelvic RT (14% vs 0%); no pelvic recurrences were observed.<sup>44</sup>

Patients with *Stage IB, grade 1 and 2* endometrioid adenocarcinoma may benefit from RT, as suggested by results of the PORTEC-1 trial. Patients treated postoperatively with pelvic RT had lower 5-year vaginal recurrence than those treated with surgery alone (1% vs 10% for grade 1 and 2% vs 13% for grade 2). Rates of pelvic recurrence within 5 years after surgery did not differ between groups, raising the question of whether pelvic RT provides greater benefit than intravaginal RT.<sup>44</sup> Two randomized trials have compared the two treatments, neither of which required pelvic lymphadenectomy. In the PORTEC-2 trial, 427 patients were randomized to pelvic ( $n = 214$ ) or intravaginal RT ( $n = 213$ ) after TAH/BSO. Inclusion criteria in this trial were more stringent

than those in PORTEC-1: Patients with Stage I were included if they were 60 years and older and had either grade 3 tumors with <50% myometrial invasion or grade 1 and 2 tumors with >50% invasion. Patients with endocervical mucosal involvement were also included. The 10-year vaginal recurrence rates were similar (3.4% for intravaginal RT vs 2.4% for pelvic RT,  $P = 0.55$ ), pelvic recurrence was higher after intravaginal RT (6.3% vs 0.9% for pelvic RT,  $P = .004$ ), however the 10-year isolated pelvic recurrence were not significantly different (2.5% vs 0.5%,  $P = 0.1$ ). In addition, neither OS nor disease-free survival differed significantly between the two arms.<sup>45</sup> In the other trial, reported by Sorbe et al.,<sup>46</sup> patients with Stage I endometrioid adenocarcinoma with at least one of the following risk factors were randomized to adjuvant intravaginal RT ( $n = 263$ ) or pelvic plus intravaginal RT ( $n = 264$ ): grade 3,  $\geq 50\%$  myometrial invasion, or DNA aneuploidy. Rates of vaginal recurrence within 5 years did not differ between the two arms (2.7% for intravaginal RT alone vs 1.9% for intravaginal plus pelvic RT,  $P = .555$ ). The pelvic recurrence rate, however, did differ (5.3% for intravaginal RT alone vs 0.4% for combined pelvic and intravaginal RT,  $P = .0006$ ). OS did not significantly differ between the two arms (90% vs 89%, respectively).<sup>46</sup>

The data from these trials indicate that the omission of pelvic RT in favor of intravaginal RT does not significantly increase the risk of vaginal recurrence. Although the rate of pelvic recurrence in the PORTEC-2 and Sorbe trials, where lymphadenectomy was not routinely done, was lower after pelvic RT, this is not a sufficient reason to recommend routine lymphadenectomy or pelvic RT for two reasons. First, neither pelvic RT nor lymphadenectomy has been shown to improve survival in patients with early-stage endometrial cancer.<sup>37,38</sup> Second, in the PORTEC-2 trial, most patients with pelvic recurrences had simultaneous distant metastasis.<sup>45</sup> The 5-year rate of isolated pelvic recurrence was only 1.5% in the intravaginal RT arm compared to 0.5% for pelvic RT ( $P = .3$ ). At MSKCC, most patients with Stage IB, grade 1 or 2 endometrial cancer undergo SLN mapping, and if the nodes are pathologically negative, they receive intravaginal RT alone.

Patients with *Stage IB, grade 3* endometrioid adenocarcinoma were not enrolled in the PORTEC-1 or PORTEC-2 trials because omitting pelvic RT when lymphadenectomy was not performed was not justifiable. In the registry study reported by Creutzberg et al.,<sup>44</sup> 99 patients with  $\geq 50\%$  myometrial invasion and grade 3 disease were treated with postoperative pelvic RT. The 5-year rates of vaginal recurrence, pelvic recurrence, and distant relapse were 5%, 8%, and 31%, respectively. Very few investigators would recommend surgery alone for these patients. In fact, an argument could be made that pelvic RT might be needed even after a negative lymphadenectomy, especially for older patients and those with LVI. In the GOG-99 trial, 190 patients with Stage I or II disease (excluding those without myometrial invasion) underwent TAH/BSO, pelvic washing, and pelvic/para-aortic lymph nodes sampling and then were randomized to observation versus pelvic RT.<sup>47</sup> The 2-year rate of relapse was significantly lower in patients who received adjuvant pelvic RT (3% vs 12%,  $P = .007$ ). However, 4-year OS did not differ (92% for RT vs 86% for surgery alone,  $P = .557$ ). The benefit of pelvic RT was predominantly seen in a subset of patients designated “high-intermediate risk (HIR)” based on advancing age, grade 2 or 3 tumors, outer third myometrial invasion, and LVI. Among those receiving pelvic RT, the overall death rate was somewhat lower compared with those on the observation arm (HR, 0.73; 90% confidence interval [CI], 0.43 to 1.26) in the HIR subgroup.<sup>47</sup> At MSKCC, patients with >50% myometrial invasion and grade 3 disease who are HIR per GOG-99 would be offered postoperative pelvic RT even in the setting of negative lymphadenectomy.

Patients with *endocervical mucosal invasion* are no longer considered Stage II per the FIGO 2009 staging system. This subset of patients represented only 2.3% of patients in NRG Oncology/GOG-120.<sup>18</sup> Although they are considered Stage IA, there are no data to indicate that observation is the optimal treatment. In fact,

such patients were randomized to either pelvic RT or intravaginal RT in the PORTEC-2 trial.<sup>45</sup> At MSKCC, these patients are treated with postoperative intravaginal RT. For patients with *Stage II* disease (cervical stromal invasion), omitting pelvic RT may be considered, in the setting of negative lymphadenectomy, based on reported low rates of vaginal and pelvic recurrence for patients who undergo lymphadenectomy and intravaginal RT.<sup>48</sup> However, patients in these series were highly selected. At MSKCC, intravaginal RT alone, as opposed to pelvic RT, is considered only for select *Stage II* patients who meet the following criteria: grade 1 and 2 endometrioid histology, depth of cervical stromal invasion of <50%, and adequate lymphadenectomy and SLN mapping. Patients with *Stage II*, grade 3 or those with cervical stromal invasion >50% are treated with pelvic RT.

In the PORTEC-2 trial, the long-term health-related quality of life (HRQL) analysis at 7 years showed that pelvic RT is associated with more long-term bowel symptoms than intravaginal RT. Rates of clinically relevant fecal leakage were 10.6% for pelvic RT and 1.8% for intravaginal RT ( $P = .03$ ), rates of diarrhea were 8.4% and 0.9% ( $P = .04$ ), rates of limitations due to bowel symptoms were 10.5% and 1.8% ( $P = .001$ ), and rates of bowel urgency were 23.3% and 6.6% ( $P < .001$ ), respectively. Urinary urgency was reported by 39.3% of pelvic RT patients and 25.5% of intravaginal RT patients ( $P = .05$ ). No difference in sexual activity was seen between treatment arms.<sup>49</sup>

In summary, if postoperative radiation is needed in early-stage endometrioid adenocarcinoma, the preferred approach is intravaginal RT rather than pelvic RT in most cases. At MSKCC, intravaginal RT is given as an outpatient treatment using high-dose-rate iridium-192 (18 Gy to 21 Gy given in three fractions with 1 to 2 weeks between fractions). Intravaginal RT targets the most likely site of recurrence in patients with early-stage disease (approximately 80% of recurrences are vaginal), and its morbidity profile is better than pelvic RT (Fig. 50.4).

### Chemotherapy

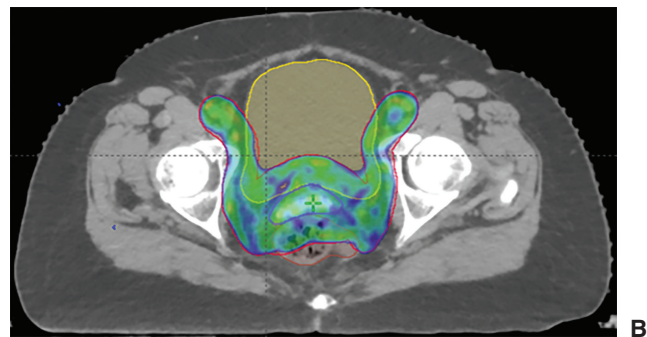
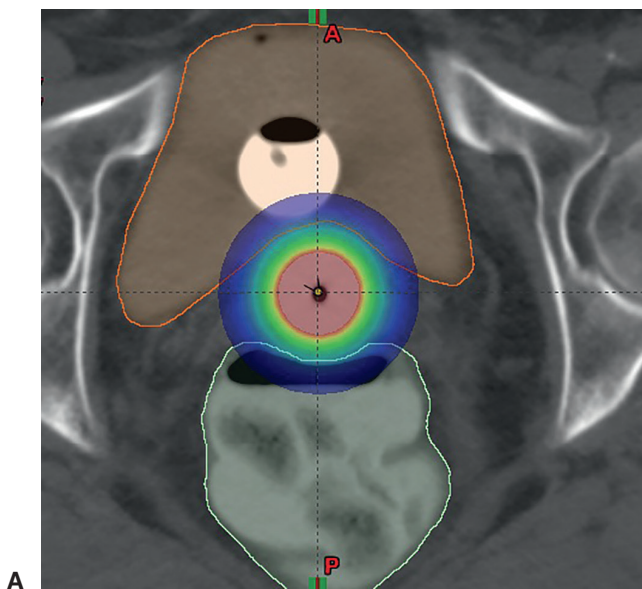
Adjuvant *chemotherapy* is not routinely recommended for patients with *Stage I* or *II* endometrioid adenocarcinoma. Two recent prospective randomized trials compared chemoradiation to radiation in early-stage, high-risk endometrial cancer. The first is the GOG-249 trial, where 601 patients with *Stage I* or *II*, HIR endometrioid adenocarcinoma and all *Stage I* or *II* serous or clear cell carcinoma were randomized to either pelvic RT (45 Gy) or intravaginal RT

plus three cycles of carboplatin plus paclitaxel.<sup>50</sup> At a median follow-up of 53 months, no significant differences were observed in 5-year recurrence-free survival (RFS) (76% in both arms), OS (87% for pelvic RT vs 85% for chemotherapy and intravaginal RT), or rates of vaginal relapse or distant metastasis. Patients who received intravaginal RT plus chemotherapy had a higher rate of pelvic/para-aortic recurrence (9% vs 4%) and experienced greater toxicity. The second trial was PORTEC-3, which included 660 patients with high-risk early-stage and *Stage III* disease (endometrioid, serous, and clear cell) randomized to either pelvic RT (46.8 Gy) or pelvic RT with concurrent cisplatin (50 mg/m<sup>2</sup>) on weeks 1 and 4 followed by four cycles of carboplatin plus paclitaxel.<sup>51</sup> In the subset of patients ( $n = 365/660$ ) with *Stage I* or *II* disease, neither 5-year OS nor RFS differed between the treatment arms (OS, 83% for chemoradiation vs 82% for RT; HR, 0.84 [95% CI, 0.52 to 1.38;  $P = .5$ ]; RFS, 81.3% vs 77.3%; HR, 0.87 [95% CI, 0.56 to 1.36;  $P = .54$ ]).

Chemotherapy is more widely used for patients with serous than for those with endometrioid histology. PORTEC-3 showed a benefit of chemotherapy for the 105 patients with serous (*Stage I* to *III*) treated with chemoradiation compared to radiation in terms of 5-year RFS rates (59.7% vs 47.9%;  $P = .008$ ) and 5-year overall survival (71.4% vs 52.8%;  $P = 0.037$ ).<sup>51</sup> In a study from MSKCC, 41 patients with *Stage I* or *II* serous cancer (excluding those with no residual disease in the hysterectomy specimen) were treated with postoperative intravaginal RT and six cycles of paclitaxel plus carboplatin, 85% of whom completed treatment as prescribed. At a median follow-up of 58 months, the 5-year RFS and OS rates were 85% and 90%, respectively. The 5-year actuarial recurrence rates were 9% in the pelvis, 5% in the para-aortic nodes, and 10% at distant sites. None of the patients developed vaginal recurrence.<sup>52</sup> Management of early stage clear cell carcinoma often mimic that for serous, yet it is not clear if adjuvant chemotherapy is warranted across the board.<sup>53</sup>

### Treatment of Stage III Disease

The optimal form of adjuvant therapy in patients with *Stage III* endometrial cancer continues to be controversial. Two randomized trials compared adjuvant external-beam radiation (EBRT) to chemoradiation. The first trial, conducted by the MaNGO ILI-ADE III study (approximately two-thirds of patients had *Stage III* disease), did not demonstrate significant improvement in outcome.<sup>54</sup> The second trial, PORTEC-3, showed an RFS advantage



**Figure 50.4** **A:** Intravaginal radiation therapy conformal dose distribution, **B:** Axial pelvic IMRT dose distribution.

for chemoradiation in the subset of patients ( $n = 295$ ) with Stage III disease (5-year RFS of 70.9% vs 58.4%;  $P = .011$ ) and in OS (78.5% chemoradiation vs 68.5% RT,  $P = .043$ ).<sup>51</sup>

Only one trial (GOG-258) compared chemoradiation to chemotherapy alone. In that study, 736 patients with Stage III and IVA or Stage I or II serous or clear cell endometrial cancer with positive cytology were randomized to six cycles of carboplatin plus paclitaxel or postoperative external-beam RT with concurrent cisplatin (50 mg/m<sup>2</sup>) on weeks 1 and 4 followed by four cycles of carboplatin plus paclitaxel.<sup>55</sup> At a median follow-up of 47 months, there was no difference in 5-year RFS (59% chemoradiation vs 58% in chemotherapy-only arm). Overall survival rates were not reported. Chemoradiation was associated with lower rates of vaginal recurrence (2% vs 7%) and pelvic/para-aortic recurrence (11% vs 20%) but more distant recurrence (27% vs 21%).

The lack of a clear conclusion from these randomized trials is due in large part to the heterogeneity of this group of patients. In a report from MSKCC on 192 patients with Stage III (FIGO 2009) endometrial cancer, the following three factors emerged as independent predictors of relapse and death from endometrial cancer: >50% myometrial invasion, positive peritoneal cytology, and aggressive histology, defined as grade 3 endometrioid, serous, clear cell, or undifferentiated.<sup>56</sup> The 5-year relapse rates were 13% for patients with no risk factors, 27% for patients with one risk factor, and 62% for patients with two or more risk factors ( $P < .001$ ). The corresponding 5-year rates of death from endometrial cancer were 11%, 20%, and 56% ( $P < .001$ ). Therefore, for patients with Stage III disease with two or more risk factors, especially positive peritoneal cytology, full systemic chemotherapy (six cycles of paclitaxel and carboplatin) is preferred over chemoradiation. For those with only one risk factor (e.g., lymph node involvement only), using chemoradiation similar to PORTEC-3 seems reasonable.

Emerging data from PORTEC-3 trial seem to indicate that the benefit of chemoradiation differed based on the molecular profile of the tumor.<sup>57</sup> The largest benefit was seen in patients with p53abn (5-year RFS was 59% vs 36%;  $P = 0.019$  in favor of chemoradiation), whereas for MMRd tumor the rate was worse for chemoradiation (68% vs 76%;  $P = 0.428$ ). For tumor with POLEmut, the outcome was excellent for both arms (100% chemoradiation vs 97% radiation). For those designated as NSMP, the 5-year RFS was 80% for chemoradiation compared to 68% for radiation ( $P = 0.234$ ).

Another area of controversy is how best to manage patients with early-stage endometrial cancer who were found to have isolated tumor cells (ITCs) on SLN mapping, i.e., <200 cells and <0.2 mm. In a SEER data analysis of 6472 patients, ITCs were seen in 1.7% of patients who otherwise had disease limited to the uterus after surgical staging.<sup>58</sup> Patients with ITCs were more likely to receive postoperative treatment (81.8% vs 31.7%,  $P < 0.001$ ), including EBRT alone (25.1% vs 3.2%) and chemotherapy/EBRT (16.3% vs 1.9%). Whether it is appropriate to treat patients with ITCs as stage IIIC disease is being questioned, however. In a multi-institutional study of patients with ITCs who otherwise were Stage I/II with endometrioid histology, it was found that RFS was not significantly influenced by the use of adjuvant therapy. Of the 175 patients, 43% received no adjuvant therapy or intravaginal RT only. When compared to those who received chemotherapy +/- EBRT, the RFS was not significantly different. Overall, the risk of recurrence in the retroperitoneal and/or distant site was only 4.6%, irrespective of treatment.<sup>59</sup> Clearly, more data are needed before making any definitive recommendations on patients with ITCs, but as is the case for all aspect of treatments in endometrial cancer, it is important to utilize any additional gained insights in the context of other prognostic factors.

When using radiation in Stage III endometrial cancer, the preferred approach is intensity-modulated radiation therapy (IMRT) because of its favorable morbidity profile. In a report from MSKCC, 46 patients with Stage I to III (78% Stage III) endometrial cancer were treated with postoperative IMRT. At a median follow-up of 52 months, 5-year RFS was 88% and 5-year OS was 97%. Chemotherapy

was given to 30 of the 46 patients; 5 had grade 3 leukopenia, 8 had grade 2 anemia, and 2 had grade 2 thrombocytopenia.<sup>60</sup>

In summary, patients with Stage III endometrial cancer are very heterogeneous; treatment should thus be individualized. Concurrent chemoradiation followed by four cycles of paclitaxel plus carboplatin seems to provide excellent outcomes for most patients. The preferred form of radiation at MSKCC is pelvic/extended-field IMRT to 50.4 Gy in 28 fractions. Patients who are at high risk for peritoneal relapse should be treated with full systemic therapy. If the data from PORTEC-3 molecular profiling are confirmed, it could further help optimize the type of adjuvant therapy to consider; POLEmut, perhaps RT alone, MMRd, consider immunotherapy with RT, and for P53abn and NSMP consider investigational agents +/- radiation.

## Treatment of Distant Recurrence

Patients with distant recurrence are not generally curable, and treatment choices should take both efficacy and toxicity into account, particularly because many patients with recurrent endometrial cancer are elderly.

## Endocrine Therapy

Single agent progestins have been used in the management of recurrent endometrial cancer. Results in unselected patient groups show limited activity (overall response rate of 23.3%) with short median PFS (2.9 months) and a median OS of 9.2 months.<sup>61</sup> Patients with well-differentiated tumors, typically expressing estrogen receptors (ERs) and/or progesterone receptors (PRs), had the highest chance of response to progestin therapy.

Alternating tamoxifen and medroxyprogesterone acetate (MPA) or megestrol acetate (MA), based on the hypothesis that the antiestrogen therapy could upregulate the PR, provides a response rate in the range of 30% in a chemotherapy-naïve population. In previously treated patients, the use of tamoxifen, aromatase inhibitors (such as letrozole), and luteinizing hormone-releasing hormone (LHRH) agonists has mostly yielded lower response rates than seen with progestins.<sup>61</sup>

Inhibiting the PI3K/Akt pathway could potentially reverse progestin resistance in endometrial cancer,<sup>61</sup> and attempts have been made to improve results of endocrine therapy by combining it with an mTOR inhibitor. A phase II study evaluated everolimus/letrozole/metformin limited to women with tumors of endometrioid histology and treated with one to two prior chemotherapy regimens. There were 62 patients enrolled. Median follow-up was 17.9 months. Median progression-free survival was 5.7 and OS was 19.6 months. Positive progesterone receptor expression was associated with higher clinical benefit (89.5% vs 27.3%,  $P = 0.001$ ).<sup>62</sup>

A phase II GOG trial randomly assigned women with advanced or recurrent endometrial cancer to either MPA plus tamoxifen (MT) or everolimus plus letrozole (EL). Median follow-up was 37 months. Median PFS for EL and MT arms was 6 months and 4 months, respectively. On EL arm, chemo-naïve patients demonstrated a 28-month median PFS compared to only 4 months for those who had prior chemotherapy. This suggests that EL is active in patients with chemo-naïve recurrent endometrial cancer.<sup>63</sup>

A randomized double-blind placebo-controlled phase II trial of CDK4/6 inhibitor palbociclib combined with letrozole (L) versus letrozole plus placebo in patients with estrogen receptor-positive advanced/recurrent endometrial cancer showed a significantly improved PFS with L + palbociclib compared with L + placebo: median 8.3 versus 3.0 months, respectively. Disease control rate at 24 weeks was 64% versus 38% in favor of L + palbociclib.<sup>64</sup> Further studies of CDK4/6 inhibitors plus endocrine therapy are also underway.

In summary, endocrine therapy should be considered for women with low-grade ER-positive tumors, particularly if they have a long interval from diagnosis to recurrence. Commonly used

regimens in the United States are MA 80 mg twice a day and MA 80 mg twice a day for 3 weeks alternating with tamoxifen 20 mg twice a day for 3 weeks. Letrozole in combination with everolimus and/or other targeted therapies can also be considered for endometrioid adenocarcinomas of the uterus.

## Cytotoxic Chemotherapy

**First-line Chemotherapy:** Most women with recurrent or Stage IV endometrial carcinoma should be assessed for treatment with carboplatin and paclitaxel (TC). This is based on a phase III, noninferiority trial for those with Stage III, Stage IV, or recurrent disease comparing TC to paclitaxel-doxorubicin-cisplatin (TAP). There were 1,381 patients enrolled; noninferiority of TC to TAP was concluded for OS (median, 37 vs 41 months, respectively; hazard ratio [HR], 1.002; 90% CI, 0.9 to 1.12), and PFS (median, 13 vs 14 months; HR, 1.032; 90% CI, 0.93 to 1.15). Neutropenic fever was reported in 7% of patients receiving TAP and 6% of those receiving TC. Grade > 2 sensory neuropathy was recorded in 26% of patients receiving TAP and 20% receiving TC ( $P = .40$ ). More grade  $\geq 3$  thrombocytopenia (23% vs 12%), vomiting (7% vs 4%), diarrhea (6% vs 2%), and metabolic (14% vs 8%) toxicities were reported with TAP. Neutropenia (52% vs 80%) was more common with TC. There was small but potentially meaningful difference in the quality of life assessment, favoring the TC regimen. The conclusion from that trial was TC is the global first-line standard for advanced endometrial cancer.<sup>65</sup>

The combination of liposomal doxorubicin plus carboplatin, which produces minimal alopecia and neuropathy, might be a reasonable alternative, particularly for patients with contraindications to taxane therapy. A phase II trial of pegylated liposomal doxorubicin plus carboplatin in the front-line therapy of advanced endometrial cancer reported a response rate of 59.5% and a median survival of 80.1 weeks.<sup>66</sup>

In summary, TC therapy every 3 weeks is the standard first-line chemotherapy treatment in the United States for women with Stage IV endometrial cancer.

**Second (and Later)-line Chemotherapy:** The efficacy of second-line cytotoxic chemotherapy remains very limited. Women who have a prolonged disease-free interval after prior carboplatin plus taxane therapy can be retreated with a similar regimen. Nagao et al. reviewed 262 patients, 49% of whom had a platinum-free interval of <12 months and 51% of whom had a platinum-free interval of  $\geq 12$  months. Over half received their initial chemotherapy regimen in the adjuvant setting. They found response rates to retreatment with a platinum-based regimen for platinum-free intervals of <6 months, 6 to 11 months, 12 to 23 months, and  $\geq 24$  months to be 25%, 38%, 61%, and 65%, respectively, suggesting that the general concept of “platinum sensitivity” in selection of second-line therapy can be applied to endometrial cancer as well as ovarian cancer.<sup>67</sup> However, most trials of second-line treatments for endometrial cancer have not traditionally been prospectively stratified along these lines (which may contribute to some of the variability seen in response rates of small phase II studies in endometrial cancer to agents such as pegylated liposomal doxorubicin).

Neither doxorubicin nor platinum agents have generally produced high response rates when used as second-line chemotherapy. Results of other selected trials with standard available cytotoxic agents in patients who have received prior chemotherapy are shown in Table 50.4. Taxanes showed good activity in the days before taxane-containing therapy was the standard first-line approach. Other agents such as topotecan and gemcitabine have likewise shown minimal efficacy in previously treated populations.<sup>68–76</sup>

These histologic subtypes have been shown to have very different genetic makeup, but to date, differences in response to standard front-line cytotoxic therapy have not been observed. Specifically, histologic subtype was not identified as an independent predictor of response to doxorubicin/cisplatin-based regimens. The overall

TABLE 50.4

### Representative Single-Agent Trials in Chemotherapy-Pretreated Endometrial Cancer

| Cytotoxic Agents                                   | Dose                                           | RR               |
|----------------------------------------------------|------------------------------------------------|------------------|
| Ixabepilone <sup>68</sup>                          | 40 mg/m <sup>2</sup> q 21 d                    | 12%              |
| Oxaliplatin <sup>69</sup>                          | 130 mg/m <sup>2</sup> q 21 d                   | 13.5%            |
| Gemcitabine <sup>70</sup>                          | 800 mg/m <sup>2</sup> days 1 and 8 q 21 d      | 4%               |
| Paclitaxel <sup>71</sup>                           | 200 mg/m <sup>2</sup> q 3 wk                   | 27%              |
| Pegylated liposomal doxorubicin <sup>72</sup>      | 40 mg/m <sup>2</sup> q 28 d                    | 21%              |
| Nonpegylated liposomal doxorubicin <sup>73</sup>   | 60 mg/m <sup>2</sup> q 4 wk                    | 0%               |
| Topotecan <sup>74</sup>                            | 0.5–1.5 mg/m <sup>2</sup> /d $\times$ 5 q 21 d | 9%               |
| Docetaxel <sup>75</sup>                            | 70 mg/m <sup>2</sup> q 3 wk                    | 23%              |
| Pemetrexed <sup>76</sup>                           | 900 mg/m <sup>2</sup> q 21 d                   | 3.8%             |
| <b>Antiangiogenic Agents</b>                       |                                                |                  |
| Aflibercept <sup>77</sup>                          | 4 mg/kg IV q 14 d                              | 7%               |
| Sorafenib <sup>78</sup>                            | 400 mg PO bid                                  | 5%               |
| Sunitinib <sup>79</sup>                            | 50 mg/d PO 4 wk on 2 wk off                    | 15%              |
| Cediranib <sup>80</sup>                            | 30 mg PO qd                                    | 12.5%            |
| Cabozantinib <sup>81</sup>                         | 60 mg PO qd                                    | 11% <sup>a</sup> |
| Nintedanib <sup>82</sup>                           | 200 mg PO bid                                  | 9.4%             |
| Brivanib <sup>83</sup>                             | 800 mg PO qd                                   | 18.6%            |
| Bevacizumab <sup>84</sup>                          | 15 mg/kg IV q 3 wk                             | 13.5%            |
| <b>mTOR Inhibitors</b>                             |                                                |                  |
| Temsirolimus <sup>85</sup>                         | 25 mg IV qwk                                   | 4%               |
| Temsirolimus (no prior chemotherapy) <sup>85</sup> | 25 mg IV qwk                                   | 14%              |
| Everolimus <sup>86</sup>                           | 10 mg PO qd                                    | 0%               |
| Everolimus <sup>87</sup>                           | 10 mg PO qd                                    | 5%               |
| Ridaforolimus <sup>88</sup>                        | 12.5 mg IV qd $\times$ 5 q 14 d                | 11%              |
| Ridaforolimus <sup>89</sup>                        | 40 mg PO qd $\times$ 5 q 4 wk                  | 9%               |

<sup>a</sup>In tumors of endometrioid histology.

RR, relative risk; q, every; IV, intravenous; PO, oral; bid, twice a day; qd, every day; qwk, every week; mTOR, mammalian target of rapamycin; EGFR, epidermal growth factor receptor.

response rate to treatment was 42% (endometrioid, 44%; serous, 44%; clear cell, 32%).<sup>90</sup>

In summary, women with a prolonged disease-free interval from their front-line platinum therapy can be retreated with a platinum-containing combination. However, in general, response rates to second-line cytotoxic therapy (and noncytotoxic agents) are low.

## Immunotherapy

The combination of tumor-infiltrating lymphocytes (seen on H&E stains), PD-1/PD-L1 expression (seen on IHC), and increased mutational load (from genomic landscape) makes endometrial cancer an ideal target for immunotherapeutic interventions. The expression level of PD-1 and PD-L1 varies according to histology (40% to 80% in endometrioid, 10% to 68% in serous, and 23% to 69% in clear cell subtypes, respectively), but overall represents the highest expression among gynecologic cancers.<sup>91</sup> About 30% of primary endometrial cancers are microsatellite instability high/hypermethylated (MSI-H), and 13% to 30% of recurrent endometrial cancers are MSI-H or mismatch repair deficient (dMMR). Immune

checkpoint blockade has been explored as a therapeutic mechanism, both as monotherapy (discussed here) and in combination with other immunotherapy, targeted agents, and chemotherapy.

In a trial of the anti-PD-1 antibody, pembrolizumab, in patients with mismatch repair-deficient tumors of multiple histologies, including endometrial cancer (all of whom had received at least one prior therapy), there were 90 patients with endometrial cancer. The objective response rate was 48%, and median duration of response was not reached. Median progression-free survival was 13.1 (95% CI, 4.3 to 34.4) months, and median overall survival was not reached. Among all treated patients, 76% had  $\geq 1$  treatment-related adverse event (grades 3 to 4, 12%). There were no fatal treatment-related events. Immune-mediated adverse events or infusion reactions occurred in 28% of patients (grades 3 to 4, 7%; no fatal events). The conclusion from this study was that pembrolizumab demonstrated robust and durable antitumor activity and encouraging survival outcomes with manageable toxicity in patients with previously treated, advanced MSI-H/dMMR endometrial cancer.<sup>92</sup> Pembrolizumab is now FDA approved as monotherapy for microsatellite instability high (MSI-H) or dMMR solid tumors that have progressed following prior treatment. Pembrolizumab is also approved for treatment of advanced tumor burden high [ $\geq 10$  mutations/megabase (mut/Mb)] solid tumors. *POLE*-mutated endometrial cancers, although they do not usually recur, have even higher mutational burdens than tumors with MSI and would be expected to respond to immune checkpoint inhibition. Patients with recurrent disease should have their tumors screened for mismatch repair deficiency to evaluate their eligibility for immunotherapy.

The anti-PD-1 antibody, dostarlimab, is also FDA approved for treatment of dMMR/MSI-H endometrial cancers based on the multicenter, single-arm, open-label study. There were 104 women with deficient mismatch mutation repair endometrial cancers who were enrolled and treated with dostarlimab. There was a confirmed response in 30 patients (objective response rate, 42.3%; 95% CI, 30.6% to 54.6%); 9 patients (12.7%) had a confirmed complete response, and 21 patients (29.6%) had a confirmed partial response. Responses were durable; the median duration of response was not reached (median follow-up was 11.2 months). The estimated likelihood of maintaining a response was 96.4% at 6 months and 76.8% at 12 months. Anemia (2.9%), colitis (1.9%), and diarrhea (1.9%) were the most common grade 3 or higher treatment-related adverse events.<sup>93</sup>

Other immune checkpoint inhibitors including, nivolumab, and anti PD-L1 agents (avelumab and durvalumab) have also shown efficacy in dMMR endometrial cancers with response rates ranging from 27% to 43%.<sup>94-96</sup> All immune checkpoint inhibitors as monotherapy in microsatellite stable or MMR-proficient endometrial cancers have shown modest activity with RR ranging from 3% to 13%.<sup>91</sup>

In summary, in MSI-H or dMMR advanced endometrial cancers, PD-1 inhibitors (dostarlimab and pembrolizumab) have shown response rates of 49% and 57%, respectively, whereas PD-L1 inhibitors (avelumab and durvalumab) have shown response rates of 27% and 43%, respectively.

**Immunotherapy and Targeted Agents:** Treatment with checkpoint inhibitors, including pembrolizumab, had provided good antitumor activity in patients with microsatellite instability-high (MSI-H) or dMMR advanced endometrial carcinoma as discussed above. However, endometrial cancers that are MSI-H or dMMR represent only a fraction; furthermore, pembrolizumab monotherapy has shown less activity in patients with microsatellite-stable or mismatch repair-proficient (pMMR) disease. Therefore, to improve upon these results, combination checkpoint inhibitors and targeted therapies were investigated.

Lenvatinib, a multitargeted tyrosine kinase inhibitor of vascular endothelial growth factor receptors (VEGFR) 1 through 3, fibroblast growth factor receptors (FGFR) 1 through 4, platelet-derived growth factor receptor  $\alpha$  (PDGFR  $\alpha$ ), RET, and KIT, has

limited activity as second-line treatment for recurrent endometrial carcinoma with an objective response of 14.3%.<sup>97</sup> Lenvatinib and anti-PD1 monoclonal antibody combinations have shown superior antitumor activity compared with either monotherapy.<sup>91</sup> In a phase 1b/II study of lenvatinib plus pembrolizumab in advanced previously treated endometrial cancer, the objective response rate at 24 weeks was 38.0%, median PFS was 7.4 months, and median OS was 16.7 months.<sup>98</sup> Based on this efficacy, combination lenvatinib and pembrolizumab received accelerated approval in 2019 for advanced endometrial that is microsatellite-stable or pMMR disease. Study 309 /KEYNOTE-775 was a multicenter, open-label, randomized, controlled trial required to confirm the clinical benefit of this accelerated approval of lenvatinib plus pembrolizumab chemotherapy of physicians' choice in advanced endometrial cancer treated with at least one prior platinum-based chemotherapy regimen in any setting. In this study, among patients with advanced endometrial cancer that is microsatellite stable or pMMR, the median progression-free survival was 6.6 months for patients in the pembrolizumab and lenvatinib group and 3.8 months for those receiving investigator's choice chemotherapy ( $P < 0.001$ ). Median overall survival was 17.4 months and 12 months ( $P < 0.001$ ). Objective response rate was 30% and 15%, respectively. Median duration of response was 9.2 months and 5.7 months.<sup>99</sup> Based on these results, pembrolizumab in combination with lenvatinib received full approval for advanced endometrial carcinoma that is microsatellite stable or pMMR, who have disease progression following prior systemic therapy in any setting.

VEGF receptor tyrosine kinase cabozantinib has also been evaluated in combination nivolumab (anti-PD1) in a multi-arm trial of advanced endometrial cancers and uterine carcinosarcomas with response rates ranging from 11% to 25%.<sup>100</sup> A phase 1b/2 open label study evaluating multikinase inhibitor lucitnib in combination with anti-PD1 monoclonal antibody nivolumab in advanced solid tumors (LIO-1-NCT04042116), including endometrial cancer, is currently ongoing.

## Anti-Human Epidermal Growth Factor Receptor 2 Agents

Human epidermal growth factor receptor 2 (HER2) overexpression and/or amplification is found in 20% to 30% of serous carcinomas and a percentage of high-grade endometrioid tumors; it is rare in lower grade endometrioid tumors.<sup>24</sup> A randomized phase II trial of carboplatin-paclitaxel (control arm) versus carboplatin-paclitaxel-trastuzumab (experimental) was conducted in Her2-overexpressed uterine serous carcinomas. There were 61 patients enrolled. After a median follow-up of 25.9 months, 43 progressions and 38 deaths occurred among 58 evaluable patients. The median PFS was 8.0 months versus 12.9 months in favor of the experimental arm (HR = 0.46; 90% CI, 0.28 to 0.76;  $P = 0.005$ ). Median PFS was 9.3 months versus 17.7 months among 41 patients with Stage III to IV disease undergoing primary treatment (HR = 0.44; 90% CI, 0.23 to 0.83;  $P = 0.015$ ) and 7.0 months versus 9.2 months among 17 patients with recurrent disease (HR = 0.12; 90% CI, 0.03 to 0.48;  $P = 0.004$ ). OS was higher in the experimental arm compared with the control arm, with medians of 29.6 months versus 24.4 months (HR = 0.58; 90% CI, 0.34 to 0.99;  $P = 0.046$ ). The benefit was most notable in those with trastuzumab in Stage III to IV disease, with survival median not reached in the experimental arm versus 24.4 months in the control arm (HR = 0.49; 90% CI, 0.25 to 0.97;  $P = 0.041$ ). Toxicity was not different between arms. The conclusion from the study was that the addition of trastuzumab to carboplatin-paclitaxel improves outcomes especially in the treatment of Stage III/IV disease.<sup>101</sup>

Pertuzumab plus trastuzumab was recently evaluated in a histology-agnostic fashion in the MyPathway trial resulting in an 23.3% confirmed objective response rate, and ado-trastuzumab emtansine (TDM1) was evaluated in the Molecular Analysis for Therapy Choice (MATCH) trial revealing an objective response

rate of 5.6%.<sup>102</sup> A number of HER2 targeted therapy trials to include monoclonal antibodies, bispecific antibodies, and antibody drug conjugates are currently ongoing.

### WEE1

More than 90% of uterine serous carcinoma have TP53 mutations rendering loss of G1/S checkpoint regulation of the G2/M checkpoint by WEE1. Additionally serous carcinomas are associated with cell cycle dysregulation paired with a high level of oncogene-driven replication stress. These features make serous carcinomas sensitive to WEE1 inhibition, as recently shown in a phase II study of adavosertib, a selective oral inhibitor of the WEE1 kinase. In this study, the ORR was 29% and mDOR was 9 months.<sup>103</sup> WEE1 has emerged as an important target in serous carcinomas, and additional studies evaluating WEE1 inhibitors have been initiated.

### Antiangiogenic Agents

Angiogenesis, a normal process in the endometrium and known pathologic process in cancer, contributes to progression and metastasis in endometrial cancer.<sup>104</sup> Antiangiogenic therapies including VEGF-TKIs cediranib, cabozantinib, and sunitinib have shown activity in the treatment of endometrial cancer (see Table 50.4).<sup>77–84</sup>

Lenvatinib, a multitargeted tyrosine kinase inhibitor including VEGFR 1–3 has limited activity as a single agent for second-line treatment in recurrent endometrial carcinoma with an objective response of 14.3% as mentioned above.<sup>97</sup> Bevacizumab, an anti-vascular endothelial growth factor (VEGF) monoclonal antibody, produced a response rate of 13.5% in patients with one or two prior cytotoxic regimens, and 40% of subjects were progression free at 6 months.<sup>83</sup> A front-line randomized phase II GOG trial randomly assigned 349 chemotherapy-naïve patients to carboplatin/paclitaxel/bevacizumab, carboplatin/paclitaxel/temsirolimus, or carboplatin/ixabepilone/bevacizumab. Response rates were similar across arms (60%, 55%, and 53%, respectively). Relative to controls (GOG 209), OS duration was significantly increased in Arm 1 ( $P < 0.039$ ) but not in Arms 2 and 3; the HRs (92% CIs) were 0.71 (0.550.91), 0.99 (0.78 to 1.26), and 0.97 (0.77 to 1.23), respectively. The conclusion was PFS was not significantly increased in any experimental arm compared to historical controls.<sup>84</sup> The Multicenter Italian Trials in Ovarian and Gynecologic Malignancies (MITO) END-2 trial randomly assigned 108 women with advanced endometrial cancer who were either chemotherapy naïve or had experienced recurrence at least 6 months after prior adjuvant chemotherapy to carboplatin plus paclitaxel (CT) or carboplatin plus paclitaxel (CTB) with bevacizumab in combination and as maintenance. When comparing CT to CTB, the PFS was 10.5 versus 13.7 months (HR 0.84,  $P = 0.43$ ), overall response rate ORR 53.1% versus 74.4%, and overall survival (OS) 29.7 versus 40.0 months (HR 0.71,  $P = 0.24$ ), resulting in a nonsignificant increase in bevacizumab-treated patients. However, patients treated with bevacizumab experienced a significant increase in 6-month disease control rate (70.4% vs 90.7%). Cardiovascular events were more frequent in the experimental arm (“de novo” grade  $\geq 2$  hypertension 21% vs 0% and grade  $\geq 2$  thromboembolic events 11% vs 2% in the bevacizumab vs standard treatment arm, respectively). The conclusion was that bevacizumab combined with chemotherapy in the treatment of advanced/recurrent EC failed to demonstrate a significant increase in PFS, but preliminary data suggest some effectiveness of the antiangiogenic agent, which merits further exploration in a larger population with a better molecular characterization.<sup>105</sup>

### Phosphatidylinositol 3-Kinase/Mammalian Target of Rapamycin Pathway

*PTEN* mutations are extremely common in type I endometrial cancer. In addition, both serous and endometrioid cancers have

very high rates of other PI3K pathway alterations.<sup>24</sup> There has therefore been widespread interest in testing PI3K pathway inhibitors in endometrial cancer.<sup>61</sup>

Numerous trials involving the rapamycin analogs have been reported, with modest activity in chemotherapy-naïve patients and low activity in previously treated patients (see Table 50.4).<sup>85–89</sup> Nonetheless, sporadic patients treated with mTOR inhibitors have prolonged responses, and these responses are seen across histologic subtypes, including endometrioid, serous, and clear cell cancers.

Combinations of mTOR inhibitors with other agents are also being explored. Combinations of mTOR inhibitors with hormonal therapy and chemotherapy were described earlier (see “Endocrine Therapy” section). Two trials have combined temsirolimus with bevacizumab. In previously treated patients, the combination produced a 24.5% response rate, but there was significant toxicity, with two gastrointestinal–vaginal fistulas, one grade 3 epistaxis, two intestinal perforations, and one grade 4 thrombosis/embolism. Three patient deaths were possibly treatment related.<sup>106</sup>

A phase II trial of the allosteric AKT inhibitor, MK-2206, described a 6% response rate with no clear association between *PIK3CA* mutation and response.<sup>107</sup> Multiple trials of newer PI3K pathway agents, including pan-PI3K, dual PI3K/mTOR, AKT, and catalytic mTOR inhibitors have been completed.

### Poly (ADP-Ribose) Polymerase Inhibitors

PARP inhibitors may be a therapeutic option for some endometrial carcinomas. It has been suggested that *PTEN*, in addition to regulating the PI3K pathway, has a role in the repair of DNA damage via homologous recombination.

In a phase Ib trial, olaparib combined with the AKT inhibitor capivasertib, showed promising results with a partial response rate of 44% in the subset with endometrial cancer.<sup>108</sup> In a randomized phase II trial comparing cediranib monotherapy, olaparib monotherapy, or the combination of olaparib and cediranib the median PFS was 3.8 months for cediranib monotherapy, 2.0 months for olaparib monotherapy, and 5.5 months for the combination, and the addition of olaparib to cediranib failed to improve efficacy in advanced endometrial cancer.<sup>109</sup>

Studies are evaluating PARP inhibitors in combination with immune checkpoint inhibitors both in the front line and recurrent disease setting are ongoing.

### Treatment of Locally Recurrent Disease

Treatment for locally recurrent endometrial cancer should be individualized. Factors to consider include location of recurrence (vaginal vs pelvic) and use of prior radiation. Creutzberg et al.<sup>13</sup> reported on survival after relapse in patients with endometrial cancer who were randomized to observation versus postoperative pelvic RT (PORTEC-1 trial). In the surgery-alone arm, the 10-year survival rate was 51% for patients with vaginal recurrence and 18% for patients with pelvic recurrence. In the postoperative radiation arm, the corresponding rates were 25% and 0%, respectively.<sup>13</sup>

### Surgery

Surgery should be reserved for patients with recurrent disease in an irradiated area (i.e., received external-beam RT). Occasionally, a simple excision of a small vaginal recurrence is warranted; this is not curative but can reduce the intensity of the RT. Achieving complete gross resection is important. Moukarzel et al.<sup>110</sup> evaluated 61 patients and found age  $< 70$  at time of initial diagnosis and endometrioid histology to be associated with improved postrelapse survival ( $P = 0.008$ , 0.03, respectively).

Pelvic exenteration may be considered in women with isolated central pelvic recurrences who had prior pelvic radiation. A study by Barakat et al.<sup>111</sup> showed that the complication rate in 44 patients undergoing pelvic exenteration for recurrent endometrial cancer was 80% and the 5-year survival rate was only 20%.

## Radiation

RT is the main treatment for isolated vaginal recurrence and, to some extent, nodal recurrence. Local control is typically achieved in 40% and 70% of patients treated with salvage radiation for *isolated vaginal recurrence*. A major determinant of local control is tumor size. Wylie et al.<sup>112</sup> reported 5-year local control rates of 80% for tumors <2 cm as opposed to 54% in tumors >2 cm ( $P = .02$ ). For most vaginal recurrences, the optimal RT approach is a combination of pelvic RT and brachytherapy. The type of brachytherapy to be used in combination with pelvic RT depends on the size of vaginal recurrence. For those with disease >2 cm, interstitial brachytherapy is the preferred approach, whereas intravaginal RT could be used instead in those with smaller volume disease and especially in those who responded well to pelvic RT. The dose of radiation is typically 45 Gy with pelvic RT plus additional 20 to 30 Gy with brachytherapy. With such high doses being required, severe complications may occur, primarily related to the gastrointestinal tract and vaginal mucosa. In a study by Jhingran et al.<sup>113</sup> on 91 patients with isolated vaginal recurrence who were treated with definitive radiation (median dose, 75 Gy), the rate of grade 4 complications (requiring surgery) was 9%.<sup>113</sup> Therefore, when considering observation after surgery for early-stage endometrial cancer, the potential for such complications should be considered.

Data regarding radiation for *nodal (pelvic/para-aortic) recurrence* are more limited and less encouraging than those for vaginal recurrence. Typically, external-beam RT is used, with very limited role for brachytherapy. As mentioned earlier, in the PORTEC-1 trial, the 10-year survival rate for pelvic recurrence was only 18%.<sup>13</sup> With modern techniques such as IMRT, the salvage rate of nodal recurrence seems to be improving. Ho et al.<sup>114</sup> reported on 38 patients who had no prior external-beam RT and were treated with definitive IMRT for their regionally confined pelvic or para-aortic nodal recurrences. Additional chemotherapy was given to 33 of 38 patients. The median total dose of IMRT was 64.7 Gy. The 2-year survival was 71%, with 8% of patients experiencing grade 3 or 4 late gastrointestinal toxicity. The encouraging results of salvage radiation with newer techniques now extend to recurrent as well as oligometastatic disease.<sup>115</sup>

## Carcinosarcoma (Mixed Müllerian Tumors)

In the mixed epithelial and mesenchymal tumors category, the World Health Organization classification includes adenosarcoma and carcinosarcoma (also termed malignant mixed Müllerian tumors). Uterine carcinosarcomas are currently regarded as metaplastic carcinomas, with the sarcomatous element being derived as a result of dedifferentiation.

Uterine carcinosarcoma accounts for approximately 4.1% of uterine cancers<sup>18</sup> but accounts for a significant number of deaths from this disease. Risk factors include prior use of tamoxifen<sup>17</sup> or pelvic radiation. It is more common in African Americans.<sup>17</sup> It is rare in women younger than 40 years, with a median age of 66.8 years.<sup>24</sup> Molecular characterization of carcinosarcomas showed frequent mutations in *TP53*, *PTEN*, *PIK3CA*, *PP2R1*, *FBXW7*, and *KRAS*, similar to endometrioid and serous uterine carcinomas.<sup>116,117</sup>

Complete surgical staging is recommended because 21.1% of patients in the NRG Oncology/GOG-120 study had involvement of pelvic lymph nodes and 15.2% had para-aortic involvement.<sup>18</sup> The FIGO 2009 staging for endometrial cancer is used for carcinosarcomas. Approximately 40% of patients present with Stage III or IV disease.<sup>18</sup> Uterine carcinosarcomas are aggressive tumors with a generally poor prognosis even among patients with early-stage disease, as demonstrated in an NRG Oncology/GOG study of early-stage, high-grade endometrial cancer patients.<sup>17</sup>

Based on poor outcome, *adjuvant therapy* in the form of radiation and chemotherapy are often used. In a European Organization for Research and Treatment of Cancer (EORTC) phase III randomized trial comparing observation to adjuvant pelvic RT, 91

patients with carcinosarcoma were included. The rate of pelvic recurrence only was 4% in the pelvic RT arm compared to 24% for the surgery-alone arm, although there was no OS benefit.<sup>118</sup> GOG-150, a phase III randomized study, compared whole abdominal irradiation (WAI) to three cycles of cisplatin, ifosfamide, and mesna (CIM). Eligible patients ( $n = 206$ ) included those with Stage I to IV uterine carcinosarcoma with no more than 1 cm post-surgical residuum and/or no extra-abdominal spread. Adjusting for stage and age, the recurrence rate was 21% lower for CIM patients than for WAI patients (relative hazard, 0.789;  $P = .245$ ), and the estimated death rate was 29% lower with CIM (relative hazard, 0.712;  $P = .085$ ). Although the authors concluded that adjuvant chemotherapy does not provide a statistically significant advantage in recurrence rate or survival over WAI, the observed differences favor future trials of combination chemotherapy.<sup>119</sup>

Based on the higher rate of vaginal recurrence in the chemotherapy arm of GOG-150 (9.9% for CIM vs 3.8% for WAI),<sup>119</sup> combining intravaginal RT with chemotherapy seems warranted. In a study from MSKCC comparing the outcome of patients with Stage I or II carcinosarcoma to that of patients with serous carcinoma, there was no statistically significant difference in outcome when patients were treated with postoperative intravaginal RT and chemotherapy.<sup>120</sup> The NRG/GOG 261 is a trial in chemotherapy-naïve uterine carcinosarcoma, where patients were randomized to paclitaxel and carboplatin (PC) or to paclitaxel and ifosfamide (PI) with 3-week cycles for 6 to 10 cycles. Distribution as follows: 40% Stage I, 6% Stage II, 31% Stage III, 15% Stage IV, and 8% recurrent. The median OS was 37 versus 29 months (HR = 0.87; 90% CI, 0.70 to 1.075;  $P < .01$  for noninferiority,  $P > .1$  for superiority). The median progression-free survival was 16 versus 12 months (HR = 0.73;  $P = < 0.01$  for noninferiority,  $P < .01$  for superiority). Toxicities were similar, except that more patients in the PC arm had hematologic toxicity and more patients in the PI arm had confusion and genitourinary hemorrhage. The conclusion from that study was that PC was not inferior to the active regimen PI and should be standard treatment for UCS.<sup>121</sup>

Treatment for *relapsed disease* is noncurative. Ifosfamide, cisplatin, and paclitaxel are the only drugs studied with clear-cut activity in the therapy of carcinosarcoma. Ifosfamide produces significant toxicity, including granulocytopenia and central nervous system toxicity, in the regimens tested. Trials of doxorubicin have produced inconsistent results. Combination therapy generally produces higher response rates, but median survival is not always improved and is generally <1 year (Tables 50.5 and 50.6).<sup>78,122–131</sup> There is no evidence for effectiveness of antiangiogenic agents. Immunotherapy trials didn't include carcinosarcoma, and reports in literatures are limited to case reports or small series.<sup>132</sup>

## UTERINE SARCOMAS

Uterine sarcomas are uncommon, representing approximately 3% to 7% of all uterine cancers.<sup>133</sup> The World Health Organization classification includes endometrial stromal tumors, smooth muscle tumors, and miscellaneous mesenchymal tumors.

### Uterine Leiomyosarcoma

Leiomyosarcomas are the most common of the uterine sarcomas. They are malignant smooth muscle tumors that are believed to arise *de novo*; for most cases, evidence does not show transition from benign leiomyoma (uterine fibroids) to leiomyosarcoma. They usually present as Stage I disease with abnormal bleeding or a pelvic mass and are more common in African Americans than in Caucasians. The median age at diagnosis is in the early 50s. Approximately 0.5% of patients undergoing hysterectomy for presumed benign leiomyoma will be found to have leiomyosarcoma; distinguishing the two entities is difficult preoperatively. Because

TABLE 50.5

**Selected Phase II and III Trials of Systemic Therapy in Uterine Carcinosarcoma with No Prior Systemic Therapy**

| Trial                              | Regimen <sup>a</sup>                                                                          | No. of Patients | RR    | Median OS/<br>Comments  |
|------------------------------------|-----------------------------------------------------------------------------------------------|-----------------|-------|-------------------------|
| van Rijswijk et al. <sup>122</sup> | Cisplatin + doxorubicin + ifosfamide                                                          | 32              | 56%   | 26 mo                   |
| Sutton et al. <sup>123</sup>       | Ifosfamide 1.5 g/m <sup>2</sup> /d × 5 q 3 wk                                                 | 102             | 36%   | 7.6 mo                  |
|                                    | vs.<br>Ifosfamide + cisplatin 20 mg/m <sup>2</sup> /d × 5 q 3 wk                              | 92              | 54%   | 9.4 mo                  |
| Homesley et al. <sup>124</sup>     | Ifosfamide 2.0 g/m <sup>2</sup> /d × 3 q 3 wk                                                 | 91              | 29%   | 8.4 mo                  |
|                                    | vs.<br>Ifosfamide 1.6 g/m <sup>2</sup> /d × 3 + paclitaxel 135 mg/m <sup>2</sup> /d q 3 wk    | 88              | 45%   | 13.5 mo; <i>P</i> = .03 |
| Powell et al. <sup>125</sup>       | Carboplatin (AUC 6) + paclitaxel 175 mg/m <sup>2</sup> q 3 wk                                 | 46              | 52%   | 14.7 mo                 |
| Aghajanian et al. <sup>126</sup>   | Carboplatin (AUC 6) + paclitaxel 175 mg/m <sup>2</sup> q 3 wk + iniparib 4 mg/kg twice weekly | 17              | 23.5% | 11.3 mo                 |

<sup>a</sup>Many trials used initial dose reductions in patients with prior pelvic radiotherapy. RR, relative risk; OS, overall survival; q, every; AUC, area under the curve.

laparoscopic extraction involving morcellation risks spreading occult leiomyosarcoma throughout the abdomen, the FDA issued a “black box” warning in November 2014 regarding the use of the electromechanical morcellator device in myomectomy and hysterectomy.<sup>134,135</sup>

Leiomyosarcomas must be distinguished from atypical leiomyomas and smooth muscle tumors of uncertain malignant potential. Smooth muscle tumors of uncertain malignant potential rarely metastasize and have an intermediate histology between benign leiomyoma and leiomyosarcoma. Mitotic rate alone does not distinguish leiomyosarcomas from tumors of low or uncertain malignant potential; cytologic atypia and coagulative necrosis are important criteria for making the diagnosis.

### Treatment

Surgery including simple hysterectomy and BSO is the standard primary treatment for Stage I disease. Lymph node involvement is rare, and lymph node dissection is not required. Chest imaging at time of diagnosis is warranted, as 10% of patients will have lung metastases at time of diagnosis. A nomogram to predict survival has been developed and validated; factors included were age (older age is associated with worse prognosis), tumor size, tumor grade, cervical involvement, locoregional and distant metastases, and mitotic index.<sup>136</sup> The current FIGO (2009) staging recognizes the importance of tumor size in leiomyosarcoma.

In a EORTC phase III randomized trial comparing observation to *adjuvant pelvic RT*, 103 patients with leiomyosarcoma were included. RT produced no improvement in survival or local control (20% local recurrence rate in RT arm compared to 24% for surgery alone).<sup>118</sup>

Only limited data exist on the effect of *adjuvant chemotherapy* on survival. Published trials have been limited by small sample size, lack of control arms, and heterogeneous patient populations. An international randomized trial of adjuvant gemcitabine plus docetaxel followed by doxorubicin versus no further treatment after surgery for early-stage uterine leiomyosarcoma (uLMS), GOG-277, was closed early as a result of slow accrual.<sup>137</sup>

For patients with multifocal metastatic disease, treatment is usually *systemic chemotherapy*. The traditional front-line regimen is the combination of gemcitabine with docetaxel, which is reasonably well tolerated, although it requires the administration of granulocyte colony-stimulating factor and is associated with occasional severe pulmonary toxicity. As first-line therapy, a study of 39 women with unresectable uLMS found a 36% objective

response rate to this combination (gemcitabine at 900 mg/m<sup>2</sup> over 90 minutes on days 1 and 8, plus docetaxel at 100 mg/m<sup>2</sup> on day 8, with growth factor support).<sup>138</sup> A placebo-controlled randomized trial adding bevacizumab to gemcitabine plus docetaxel in the first-line setting did not observe better outcomes. This trial used the same dose and schedule of gemcitabine as in the earlier trials, but the docetaxel dose was reduced to 75 mg/m<sup>2</sup>, with further dose reductions of both drugs for patients with prior pelvic RT (22 of 107 patients). The objective response rate for the control arm was 31.5%, with a 6.3-month median PFS.<sup>139</sup>

The GeDDiS trial found an even lower response rate of 20%.<sup>140</sup> This trial compared six cycles of gemcitabine plus docetaxel (same schedule as in previous trials but a lower dose of 675 mg/m<sup>2</sup>) to six cycles of single-agent doxorubicin at 75 mg/m<sup>2</sup> every 3 weeks

TABLE 50.6

**Selected Phase II and III Trials of Systemic Therapy in Uterine Carcinosarcoma with Prior Systemic Therapy**

| Trial                         | Regimen <sup>a</sup>                                                                                        | No. of Patients | RR    | Median OS |
|-------------------------------|-------------------------------------------------------------------------------------------------------------|-----------------|-------|-----------|
| Curtin et al. <sup>127</sup>  | Paclitaxel 170 mg/m <sup>2</sup>                                                                            | 44              | 18.2% | n/a       |
| Miller et al. <sup>128</sup>  | Gemcitabine 600 mg/m <sup>2</sup> on days 1, 8, 15 + docetaxel 35 mg/m <sup>2</sup> on days 1, 8, 15 q 28 d | 24              | 8.3%  | 4.9 mo    |
| Miller et al. <sup>129</sup>  | Topotecan 1.5 mg/m <sup>2</sup> /d × 5                                                                      | 48              | 10%   | n/a       |
| Mackay et al. <sup>130</sup>  | Aflibercept 4 mg/kg q 2 wk                                                                                  | 19              | 0%    | 3.2 mo    |
| Nimeiri et al. <sup>78</sup>  | Sorafenib 400 mg bid                                                                                        | 16              | 0%    | 5 mo      |
| McCourt et al. <sup>131</sup> | Ixabepilone 40 mg/m <sup>2</sup> q 21 d                                                                     | 34              | 11.8% | 7.7 mo    |

<sup>a</sup>Many trials used initial dose reductions in patients with prior pelvic radiotherapy. RR, relative risk; OS, overall survival; n/a, not available; q, every; bid, twice a day.

in 257 patients with chemotherapy-naïve, high-grade soft tissue sarcoma using prospective stratification by uLMS subtype. None of the outcomes, including response rate, PFS at 24 weeks, OS, and quality of life, differed between the two treatments among all patients or within uLMS patients ( $n = 71$ ) specifically. The lower doses of gemcitabine used in this trial may have contributed to the inferior outcomes compared with other studies. However, the results suggest that doxorubicin is also an appropriate first-line therapy for metastatic uLMS.

A recent trial tested the addition of the anti-platelet-derived growth factor receptor  $\alpha$  (PDGFR) antibody olaratumab to doxorubicin in soft tissue sarcoma including uterine ( $n = 94$ ) showed no difference in outcome.<sup>141</sup>

Subsequent-line treatment potential options are listed in Table 50.7.<sup>130,142–146</sup>

Trabectedin, a drug that binds the minor groove of the DNA double helix, was shown in a large ( $n = 577$ ) randomized phase III trial to provide superior disease control when compared to dacarbazine for anthracycline-pretreated patients with locally advanced or metastatic LMS (all sites) or liposarcoma, which led to FDA approval of trabectedin in this setting. A subgroup analysis including only the 232 women with uLMS similarly found superior outcomes with trabectedin (median PFS, 4.0 vs 1.5 months;  $P = .0012$ ). Response rates were similar (11% vs 9%), as was OS (13.4 vs 12.9 months).<sup>147</sup>

Pazopanib is an orally available multikinase angiogenesis inhibitor approved by the FDA for the treatment of soft tissue sarcomas including uterine leiomyosarcoma. A phase II trial of pazopanib 800 mg by mouth daily showed a PFS at 12 weeks of 44% in patients with leiomyosarcoma treated with two or fewer prior cytotoxic regimens (all primary sites), although there was only one

partial response (PR) in a leiomyosarcoma patient (of 41 patients treated).<sup>148</sup> A subsequent randomized phase III trial (PALETTE) in pretreated patients showed a PFS of 4.6 months (all sarcoma subtypes) versus 1.6 months for placebo.<sup>149</sup>

uLMS expresses ER or PR in a significant number of cases, and hormone receptor expression has been reported to be prognostic. One study reported that only 1 of 10 early-stage PR-positive patients experienced recurrence and died, whereas 9 of 10 PR-negative patients experienced recurrence and 5 died.<sup>150</sup> Hormonal therapy has been suggested to benefit patients with advanced-stage leiomyosarcoma that expresses ER or PR, but this remains hypothetical. One prospective phase II trial treated 27 patients with leiomyosarcoma whose tumors expressed ER and/or PR with the aromatase inhibitor letrozole at a dose of 2.5 mg daily. The patients had received a median of two prior treatment regimens. There were no objective responses, but the 12-week PFS rate was 50% (90% CI, 30% to 67%), and three patients, all of whom had tumors expressing ER and PR in >90% of tumor cells, continued to receive letrozole for >24 weeks.<sup>151</sup> However, this may merely reflect the indolent natural biology of certain strongly receptor-positive uterine leiomyosarcomas.

Although uLMS has been reported to exhibit moderate expression of PD-1 (49%) and programmed cell death protein ligand 1 (PD-L1) (36%),<sup>152</sup> single-agent immune checkpoint inhibitors have not been highly successful to date. A trial of nivolumab monotherapy had no responders among the 12 uterine leiomyosarcoma patients treated.<sup>153</sup>

## Endometrial Stromal Sarcoma

Endometrial stromal sarcomas (ESSs) account for 1% of all uterine malignancies and 15% of malignant mesenchymal neoplasms of the uterus. The World Health Organization subdivides ESS into low-grade, high-grade, and undifferentiated sarcoma.

Low-grade ESSs are indolent tumors that usually occur in women aged 40 to 55 years. The tumor cells are positive for CD10 and ER or PR.<sup>133</sup> Several chromosomal translocations resulting in gene fusion have been identified in low-grade ESS, with the most common being translocation of  $t(7; 17)(p15; q21)$ , resulting in the juxtaposition of two zinc finger genes (*JAZF1-SUZ12*). In most patients, the disease is diagnosed at an early stage; 65% to 86% of low-grade ESSs are confined to the uterus at diagnosis. Estimates of the rate of lymph node metastasis range from 10% to 45%. Approximately one-third of Stage I patients eventually develop recurrent disease.<sup>133</sup> Recurrences can occur 10 to 20 years after diagnosis, and survival after recurrence may be relatively long.

High-grade ESSs have a characteristic  $t(10; 17)$  translocation that results in a *YWHAE-NUTM2A/B* fusion.<sup>133</sup> They typically express strong and diffuse cyclin D1 and KIT, have high mitotic activity (>10/10 high-power fields [HPF]) and necrosis, and may be admixed with a low-grade fibromyxoid component. Prognosis is intermediate between low-grade ESS and undifferentiated uterine sarcoma. *YWHAE* rearrangement in a histologically conventional low-grade ESS has been reported to be associated with transformation to high-grade ESS.<sup>154</sup>

## Treatment

The primary treatment for early-stage, low-grade stromal sarcoma is hysterectomy. Lymph node dissection may offer prognostic information, but it has not been shown to improve survival. In general, removal of the ovaries is recommended, as estrogen has been noted to be a proliferative stimulus. However, the data are not conclusive regarding the impact of preservation of ovaries on outcome.

Regarding the role of *adjuvant radiation*, older series have not distinguished among the ESS subtypes, and it is difficult to draw meaningful conclusions. A National Cancer Database analysis reported by Seagle et al.<sup>155</sup> suggested that use of adjuvant chemotherapy or

**TABLE 50.7**

**Selected Prospective Phase II and III Trials of Systemic Therapy in Advanced or Recurrent Uterine Leiomyosarcoma with Prior Systemic Therapy**

| Trial                                 | Regimen <sup>a</sup>                                                                                                       | No. of Patients | RR    | Median OS               |
|---------------------------------------|----------------------------------------------------------------------------------------------------------------------------|-----------------|-------|-------------------------|
| Garcia del Muro et al. <sup>142</sup> | Temozolomide 75–100 mg/m <sup>2</sup> /d                                                                                   | 11              | 45%   | n/a                     |
| Gallup et al. <sup>143</sup>          | Paclitaxel 175 mg/m <sup>2</sup> /3 h q 21 d                                                                               | 53              | 8.4%  | 12.1 mo                 |
| Look et al. <sup>144</sup>            | Gemcitabine 1,000 mg/m <sup>2</sup> /30 min on days 1, 8, 15 q 21 d                                                        | 44              | 20.5% | n/a                     |
| Hensley et al. <sup>145</sup>         | Gemcitabine 900 mg/m <sup>2</sup> over 90 min on days 1 and 8 + docetaxel 100 mg/m <sup>2</sup> on day 8 with G-CSF q 21 d | 48              | 27%   | 14.7 mo                 |
| Hensley et al. <sup>146</sup>         | Sunitinib 50 mg PO qd × 4 wk then 2 wk off                                                                                 | 23              | 8.7%  | 15.1 mo                 |
| Mackay et al. <sup>130</sup>          | Aflibercept 4 mg/kg IV q 2 wk                                                                                              | 41              | 0%    | 18.1 mo<br>17% 6 mo PFS |

<sup>a</sup>Regimens often dose reduced in patients with prior pelvic radiotherapy. RR, relative risk; OS, overall survival; n/a, not available; G-CSF, granulocyte colony-stimulating factor; PO, oral; IV, intravenous; PFS, progression-free survival.

RT is associated with decreased survival in low-grade ESS; this was considered likely to result from the likely greater (although unmeasured) extent of disease in patients receiving treatment. There are no prospective data for the use of *adjuvant hormonal therapy*, which was not associated with survival in the Seagle et al.<sup>155</sup> study. However, given its efficacy in advanced or recurrent disease, it is often recommended for women with Stage III or higher disease.

Surgical resection of limited metastases (e.g., lung) should be considered. In recurrent disease, progestins and aromatase inhibitors can both produce prolonged responses.<sup>133</sup> The role of chemotherapy is unclear. Metastatic high-grade ESS has been reported to respond to anthracycline-based and gemcitabine plus docetaxel chemotherapy.<sup>133</sup>

Undifferentiated uterine sarcoma is considered a diagnosis of exclusion. They are high-grade tumors with no defining genetic rearrangement, are less common than ESSs, and usually occur in postmenopausal women.<sup>133</sup> Half of patients present with Stage IV disease. Progression after resection of early-stage disease may occur within months. Surgery is the primary therapy for early-stage disease; as with other ESSs, the need for lymph node dissection is uncertain. Gemcitabine/docetaxel-, doxorubicin-, and ifosfamide-based regimens have been reported to yield short-lived responses.<sup>133</sup> A National Cancer Database review that considered undifferentiated endometrial sarcomas together with high-grade ESSs suggested that adjuvant chemotherapy and RT were associated with increased survival.<sup>155</sup>

## REFERENCES

1. Siegel RL, Miller KD, Fuchs H, et al. Cancer statistics, 2022. *CA Cancer J Clin* 2022;72(1):7–33.
2. Lu KH, Broadbent RR. Endometrial cancer. *N Engl J Med* 2020;383(21):2053–2064.
3. Clarke M, Devesa S, Harvey S, et al. Hysterectomy-corrected uterine corpus cancer incidence trends and differences in relative survival reveal racial disparities and rising rates of nonendometrioid cancers. *J Clin Oncol* 2019;37(22):1895–1908.
4. Hopkins BD, Goncalves MD, Cantley LC. Obesity and cancer mechanisms: Cancer metabolism. *J Clin Oncol* 2016;34(35):4277–4283.
5. Makker V, MacKay H, Ray-Coquard I, et al. Endometrial cancer. *Nat Rev Dis Primers* 2021;7(1):88. doi: 10.1038/s41572-021-00324-8.
6. Tan MH, Mester JL, Ngeow J, et al. Lifetime cancer risks in individuals with germline PTEN mutations. *Clin Cancer Res* 2012;18(2):400–407.
7. Goldstein RB, Bree RL, Benson CB, et al. Evaluation of the woman with postmenopausal bleeding: Society of Radiologists in Ultrasound-Sponsored Consensus Conference statement. *J Ultrasound Med* 2001;20(10):1025–1036.
8. Faria SC, Devine CE, Rao B et al. Imaging and staging of endometrial cancer. *Semin Ultrasound CT MRI* 2019;40(4):287–294.
9. Freidman S, Itani M, Dehdashti R. PET imaging in gynecologic malignancies. *Radiol Clin North Am* 2021;59(5):813–833.
10. Todo Y, Sakuragi N, Nishida R, et al. Combined use of magnetic resonance imaging, CA 125 assay, histologic type, and histologic grade in the prediction of lymph node metastasis in endometrial carcinoma. *Am J Obstet Gynecol* 2003;188(5):1265–1272.
11. Trimble CL, Kauderer J, Zaino R, et al. Concurrent endometrial carcinoma in women with a biopsy diagnosis of atypical endometrial hyperplasia: A Gynecologic Oncology Group study. *Cancer* 2006;106(4):812–819.
12. Felix AS, Brasky TM, Cohn DE, et al. Endometrial carcinoma recurrence according to race and ethnicity: An NRG Oncology/Gynecologic Oncology Group 210 study. *Int J Cancer* 2018;142(6):1102–1115.
13. Creutzberg CL, Nout RA, Lybeert ML, et al. Fifteen-year radiotherapy outcomes of the randomized PORTEC-1 trial for endometrial carcinoma. *Int J Radiat Oncol Biol Phys* 2011;81(4):e631–e638.
14. Alektiar KM, Venkatraman E, Abu-Rustum N, et al. Is endometrial carcinoma intrinsically more aggressive in elderly patients? *Cancer* 2003;98(11):2368–2377.
15. Luo LY, Aviki EM, Lee A, et al. Socioeconomic inequality and omission of adjuvant radiation therapy in high-risk, early-stage endometrial cancer. *Gyn Oncol* 2021;161(2):463–469.
16. Barlin JN, Soslow RA, Lutz M, et al. Redefining stage I endometrial cancer: Incorporating histology, a binary grading system, myometrial invasion, and lymph node assessment. *Int J Gynecol Cancer* 2013;23(9):1620–1628.
17. Fader AN, Java J, Tenney M, et al. Impact of histology and surgical approach on survival among women with early-stage, high-grade uterine cancer: An NRG Oncology/Gynecologic Oncology Group ancillary analysis. *Gynecol Oncol* 2016;143(3):460–465.
18. Creasman WT, Ali S, Mutch DG, et al. Surgical-pathological findings in type 1 and 2 endometrial cancer: An NRG Oncology/Gynecologic Oncology Group study on GOG-210 protocol. *Gynecol Oncol* 2017;145(3):519–525.
19. Mariani A, Dowdy SC, Keeney GL, et al. Predictors of vaginal relapse in stage I endometrial cancer. *Gynecol Oncol* 2005;97(3):820–827.
20. Bosse T, Peters EE, Creutzberg CL, et al. Substantial lymph-vascular space invasion (LVSI) is a significant risk factor for recurrence in endometrial cancer: A pooled analysis of PORTEC 1 and 2 trials. *Eur J Cancer* 2015;51(13):1742–1750.
21. Croog VJ, Abu-Rustum NR, Barakat RR, et al. Adjuvant radiation for early stage endometrial cancer with lymphovascular invasion. *Gynecol Oncol* 2008;111(1):49–54.
22. Abu-Rustum NR, Zhou Q, Gomez JD, et al. A nomogram for predicting overall survival of women with endometrial cancer following primary therapy: Toward improving individualized cancer care. *Gynecol Oncol* 2010;116(3):399–403.
23. Creutzberg CL, van Stiphout RG, Nout RA, et al. Nomograms for prediction of outcome with or without adjuvant radiation therapy for patients with endometrial cancer: A pooled analysis of PORTEC-1 and PORTEC-2 trials. *Int J Radiat Oncol Biol Phys* 2015;91(3):530–539.
24. Kandath C, Schultz N, Cherniak AD, et al. Integrated genomic characterization of endometrial carcinoma. *Nature* 2013;497(7447):67–73.
25. Mitric C, Bernardini MQ. Endometrial cancer: Transitioning from histology to genomics. *Curr Oncol* 2022;29(2):741–757.
26. Vermij L, Smit V, Nout R, et al. Incorporation of molecular characteristics into endometrial cancer management. *Histopathology* 2020;76(1):52–63.
27. Raffone A, Travaglini A, Mascolo M, et al. TCGA molecular groups of endometrial cancer: Pooled data about prognosis. *Gynecol Oncol* 2019;155(2):374–383.
28. DeLair DF, Burke KA, Selenica P, et al. The genetic landscape of endometrial clear cell carcinomas. *J Pathol* 2017;243(2):230–241.
29. Pecorelli S. Revised FIGO staging for carcinoma of the vulva, cervix, and endometrium. *Int J Gynaecol Obstet* 2009;105(2):103–104.
30. Abu-Rustum NR, Zhou Q, Iasonos A, et al. The revised 2009 FIGO staging system for endometrial cancer: Should the 1988 FIGO stages IA and IB be altered? *Int J Gynecol Cancer* 2011;21(3):511–516.
31. Milgrom SA, Kollmeier MA, Abu-Rustum NR, et al. Positive peritoneal cytology is highly predictive of prognosis and relapse patterns in stage III (FIGO 2009) endometrial cancer. *Gynecol Oncol* 2013;130(1):49–53.
32. Scott SA, van der Zanden C, Cai E, et al. Prognostic significance of peritoneal cytology in low-intermediate risk endometrial cancer. *Gynecol Oncol* 2017;145(2):262–268.
33. Schultheis AM, Ng CK, De Filippo MR, et al. Massively parallel sequencing-based clonality analysis of synchronous endometrioid endometrial and ovarian carcinomas. *J Natl Cancer Inst* 2016;108(6):djv427.
34. Walker JL, Piedmonte MR, Spirtos NM, et al. Laparoscopy compared with laparotomy for comprehensive surgical staging of uterine cancer: Gynecologic Oncology Group Study LAP2. *J Clin Oncol* 2009;27(32):5331–5336.
35. Walker JL, Piedmonte MR, Spirtos NM, et al. Recurrence and survival after random assignment to laparoscopy versus laparotomy for comprehensive surgical staging of uterine cancer: Gynecologic Oncology Group LAP2 Study. *J Clin Oncol* 2012;30(7):695–700.
36. Wright JD, Havrilesky LJ, Cohn DE, et al. Estimating potential for savings for low risk endometrial cancer using the Endometrial Cancer Alternative Payment Model (ECAP): A companion paper to the Society of Gynecologic Oncology Report on the Endometrial Cancer Alternative Payment Model. *Gynecol Oncol* 2018;149(2):241–247.
37. Benedetti Panici P, Basile S, Maneschi F, et al. Systematic pelvic lymphadenectomy vs. no lymphadenectomy in early-stage endometrial carcinoma: Randomized clinical trial. *J Natl Cancer Inst* 2008;100(23):1707–1716.
38. Kitchener H, Swart AM, Qian Q, et al. Efficacy of systematic pelvic lymphadenectomy in endometrial cancer (MRC ASTEC trial): A randomised study. *Lancet* 2009;373(9658):125–136.
39. Ballester M, Dubernard C, Lécure F, et al. Detection rate and diagnostic accuracy of sentinel-node biopsy in early stage endometrial cancer: A prospective multicentre study (SENTI-ENDO). *Lancet Oncol* 2011;12(5):469–476.
40. Rossi EC, Kowalski LD, Scalici J, et al. A comparison of sentinel lymph node biopsy to lymphadenectomy for endometrial cancer staging (FIRE trial): A multicentre, prospective, cohort study. *Lancet Oncol* 2017;18(3):384–392.
41. Barlin JN, Khoury-Collado F, Kim CH, et al. The importance of applying a sentinel lymph node mapping algorithm in endometrial cancer staging: Beyond removal of blue nodes. *Gynecol Oncol* 2012;125(3):531–535.
42. Soliman PT, Westin SN, Dioun S, et al. A prospective validation study of sentinel lymph node mapping for high-risk endometrial cancer. *Gynecol Oncol* 2017;146(2):234–239.
43. Sorbe B, Nordström B, Mäenpää J, et al. Intravaginal brachytherapy in FIGO stage I low-risk endometrial cancer: A controlled randomized study. *Int J Gynecol Cancer* 2009;19(5):873–878.